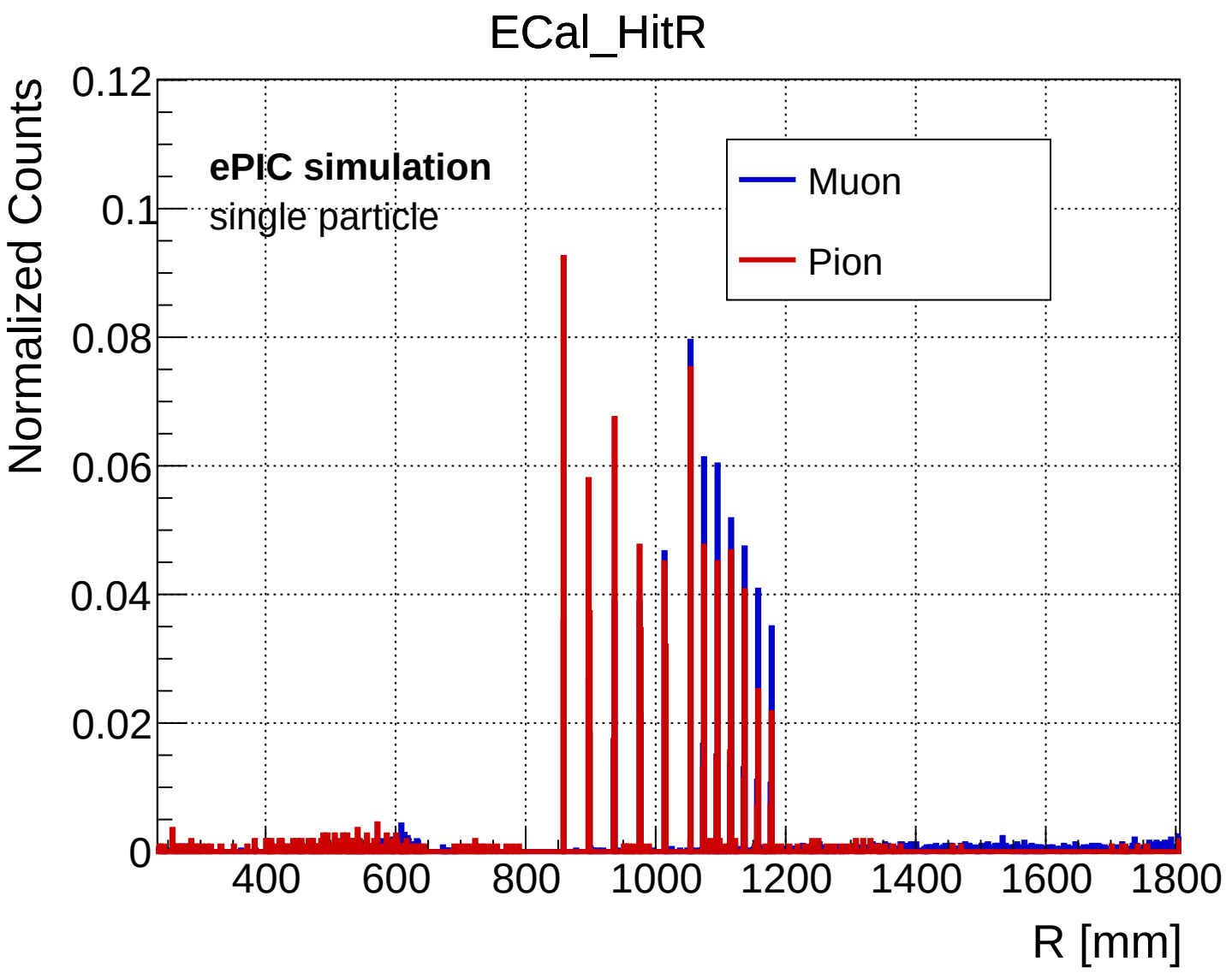
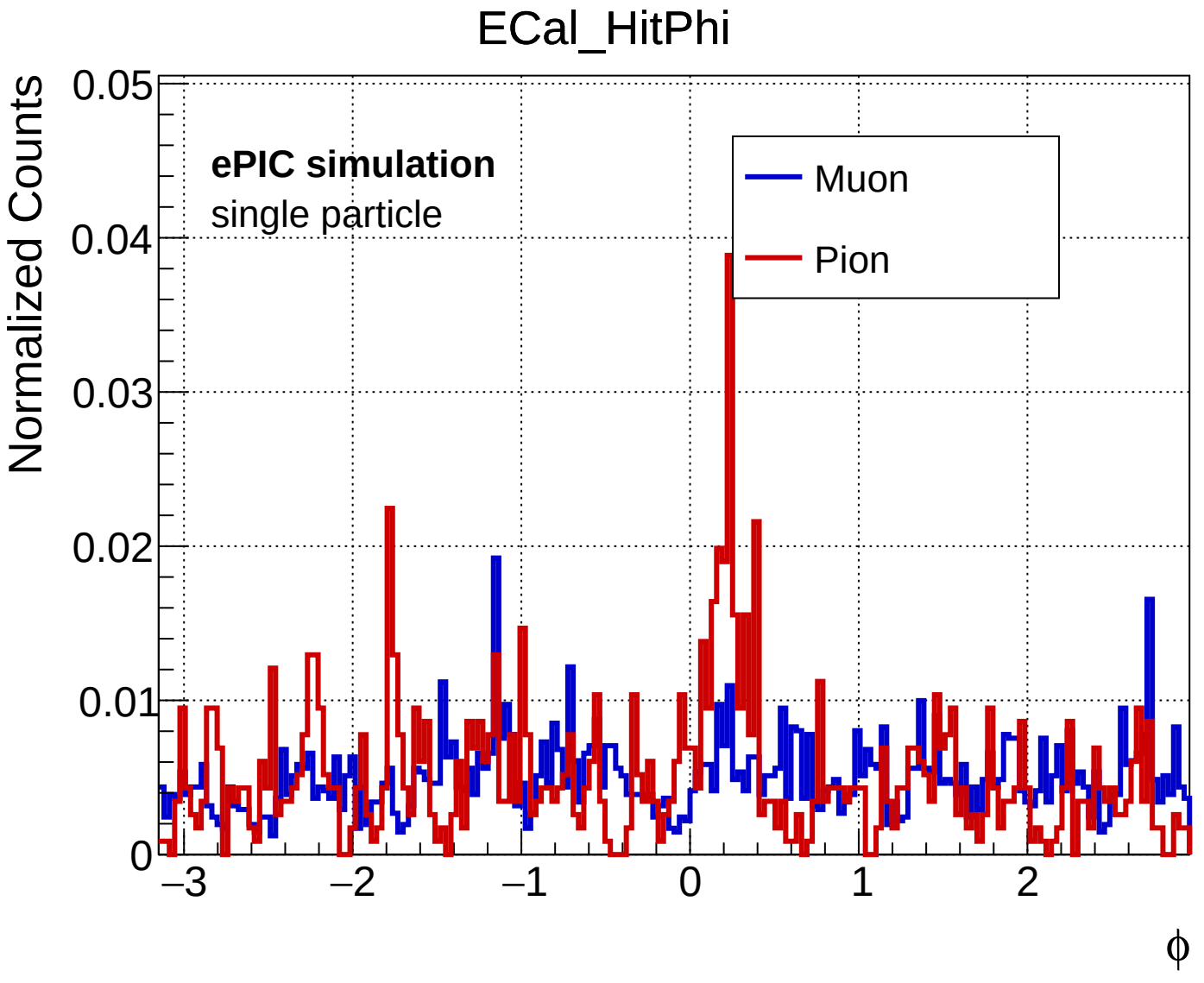
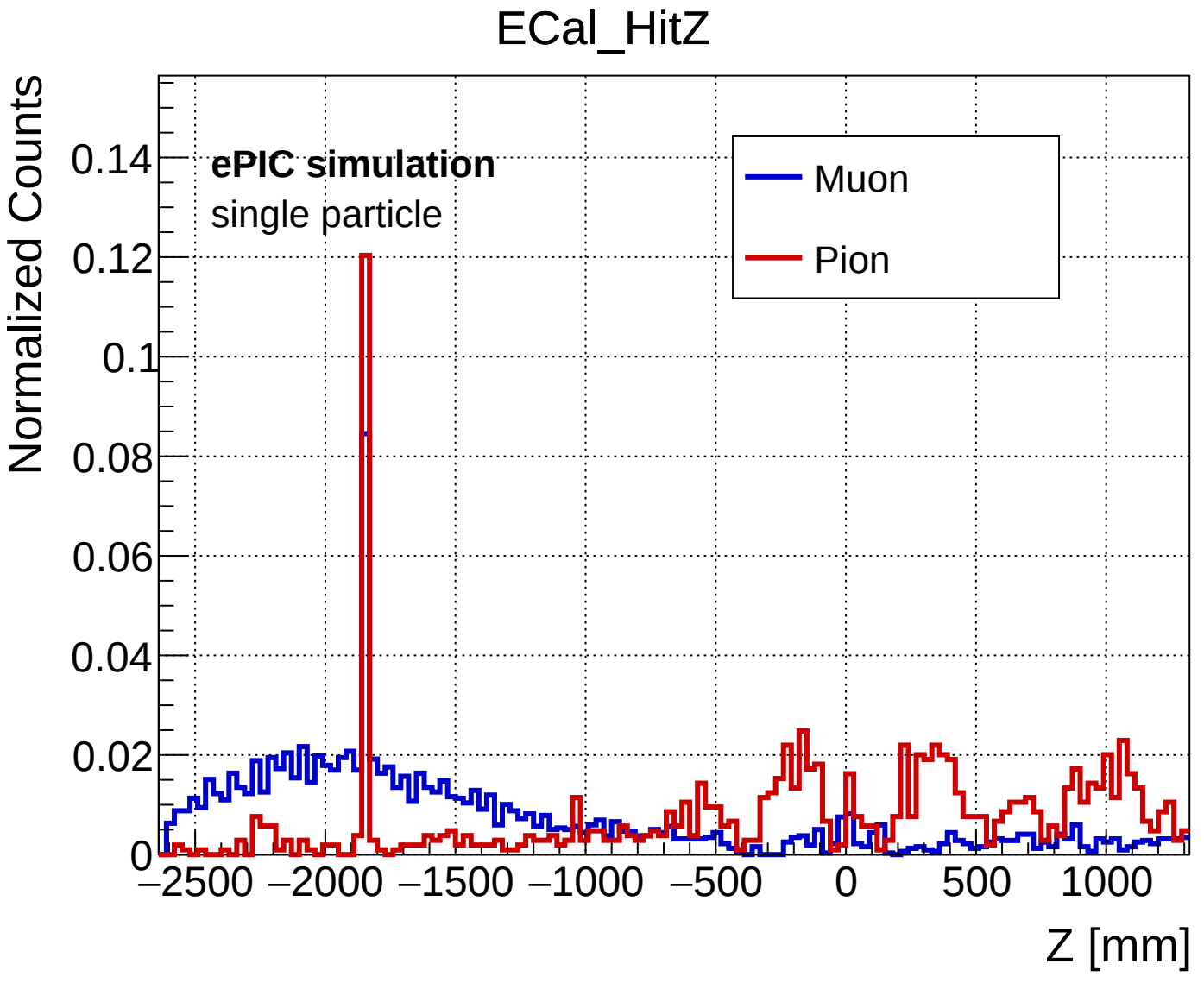
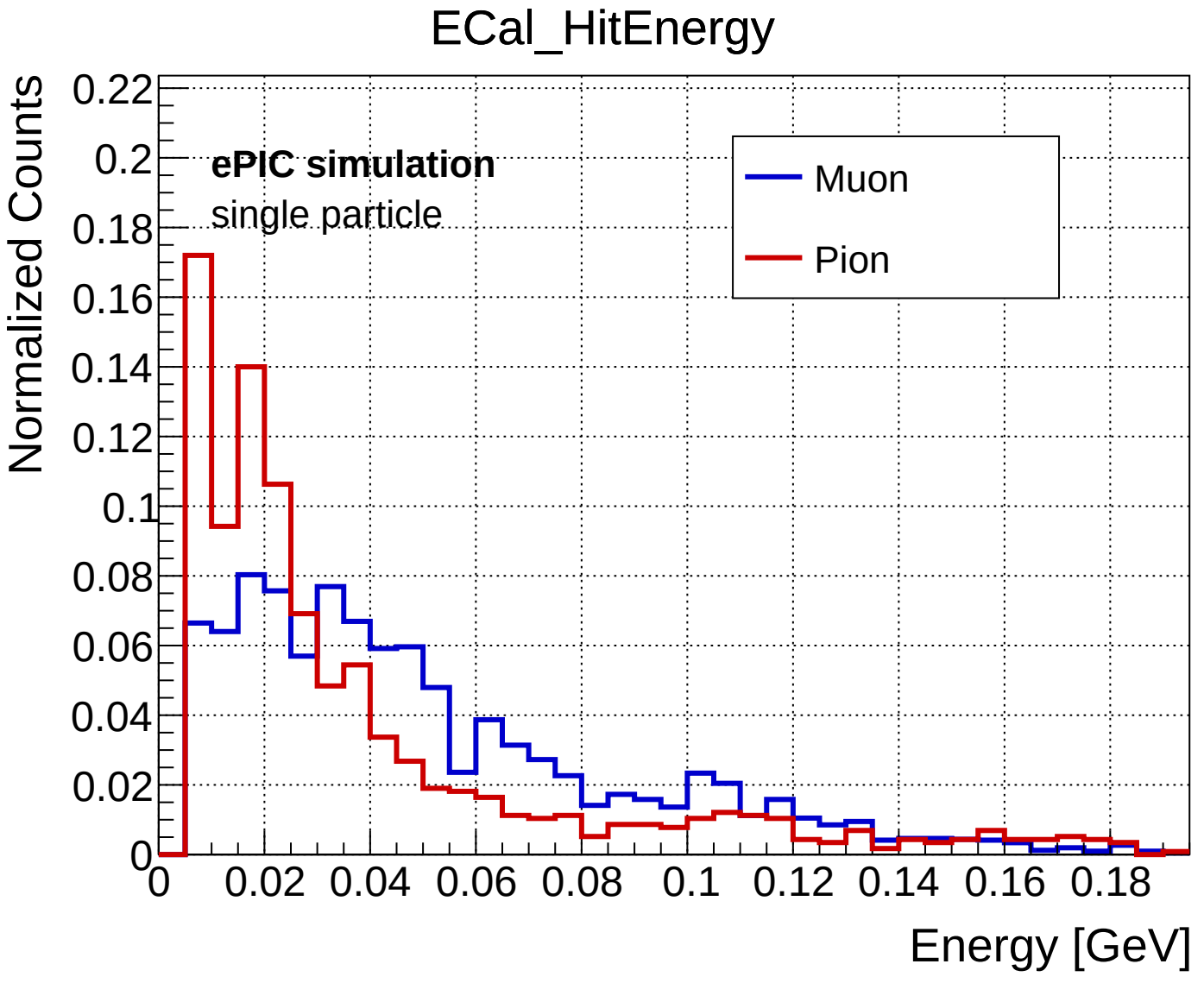
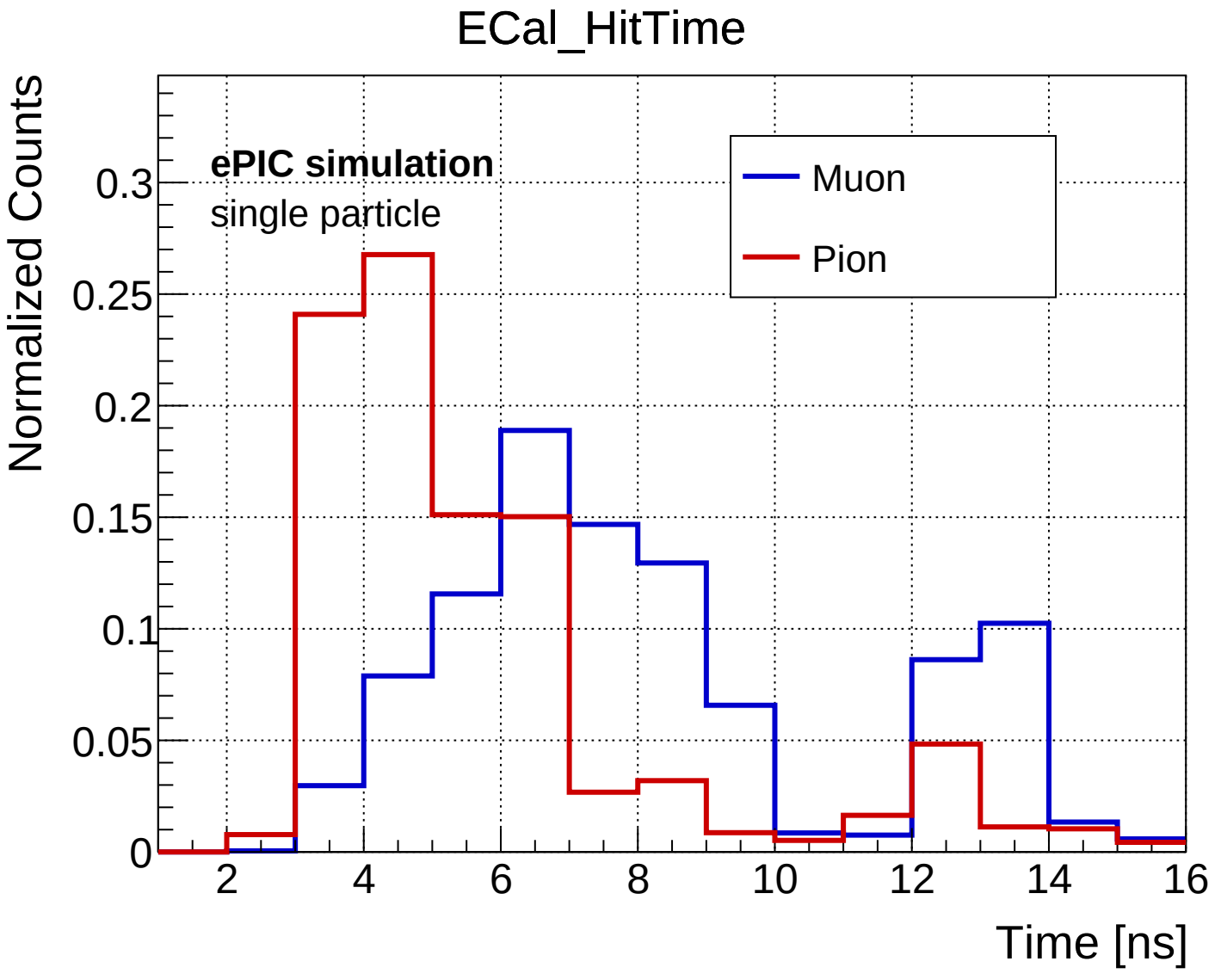
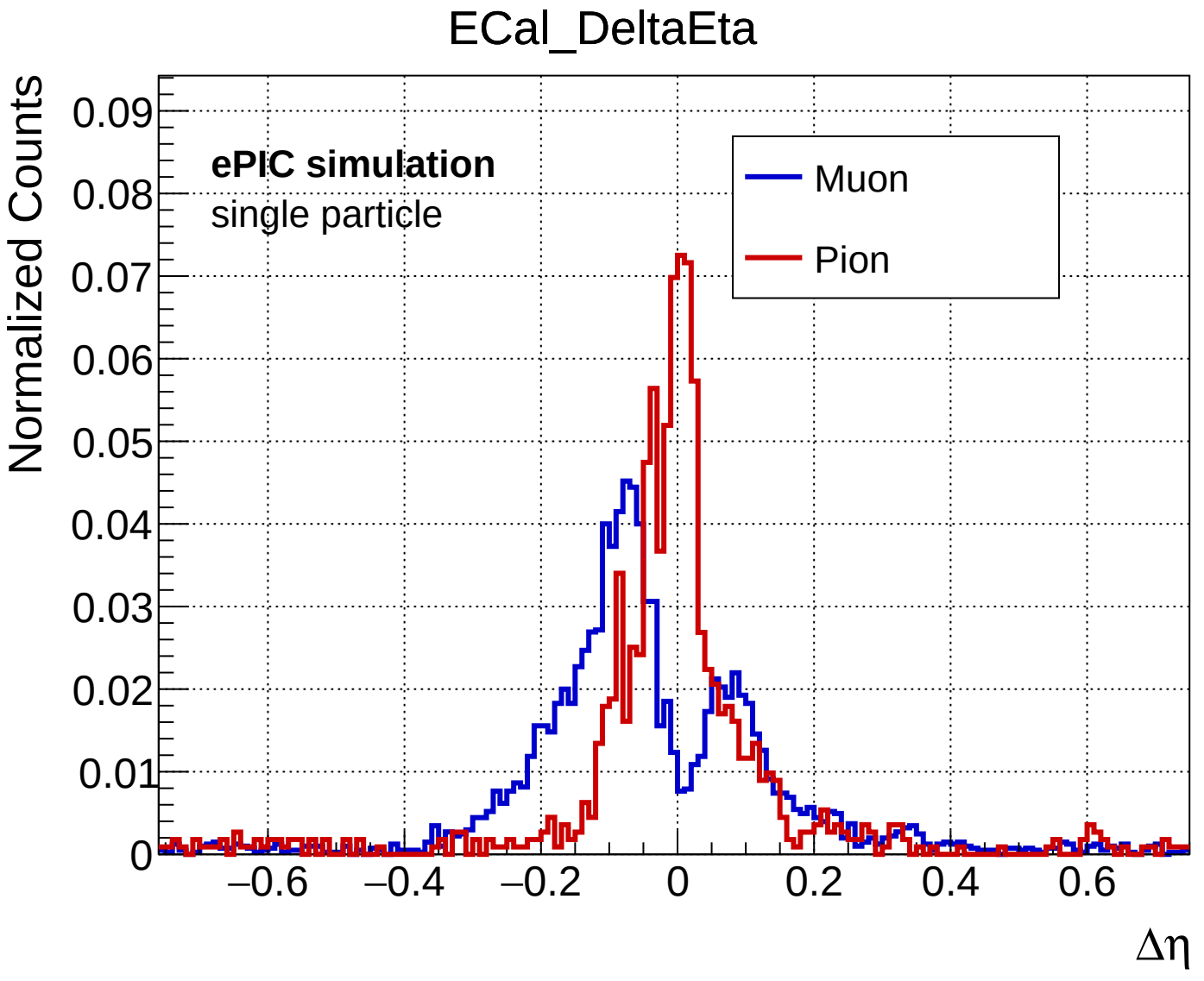

ECal_HitR
Normalized Counts
0.12
0.1
0.08
0.06
0.04
0.02
0
ePIC simulation
single particle
Muon
Pion
400
600
800
1000
1200
1400
1600
1800
R [mm]

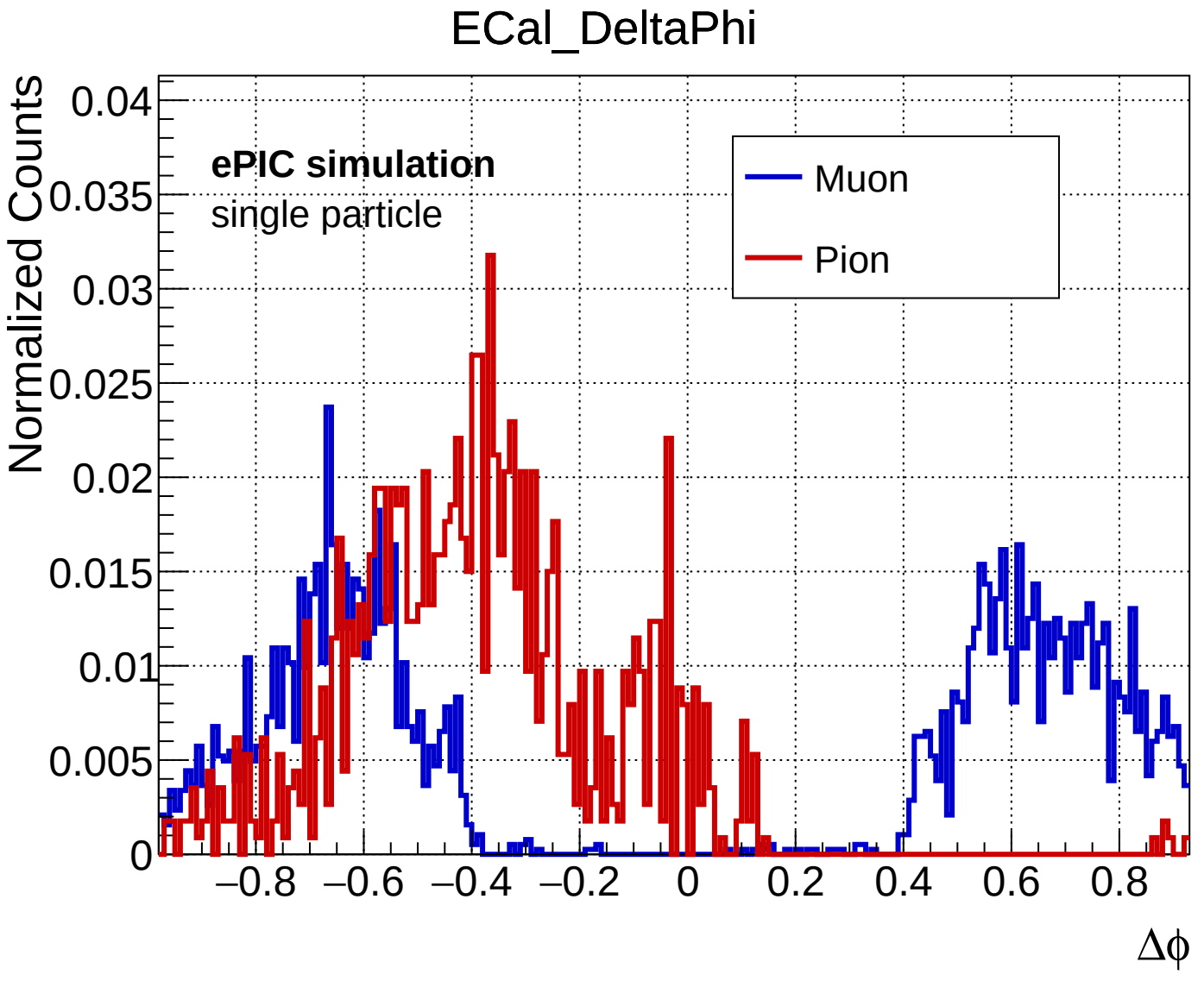

ECal_HitPhi
Normalized Counts
0.05
0.04
0.03
0.02
0.01
0
ePIC simulation
single particle
Muon
Pion
−3
−2
−1
0
1
2
ϕ

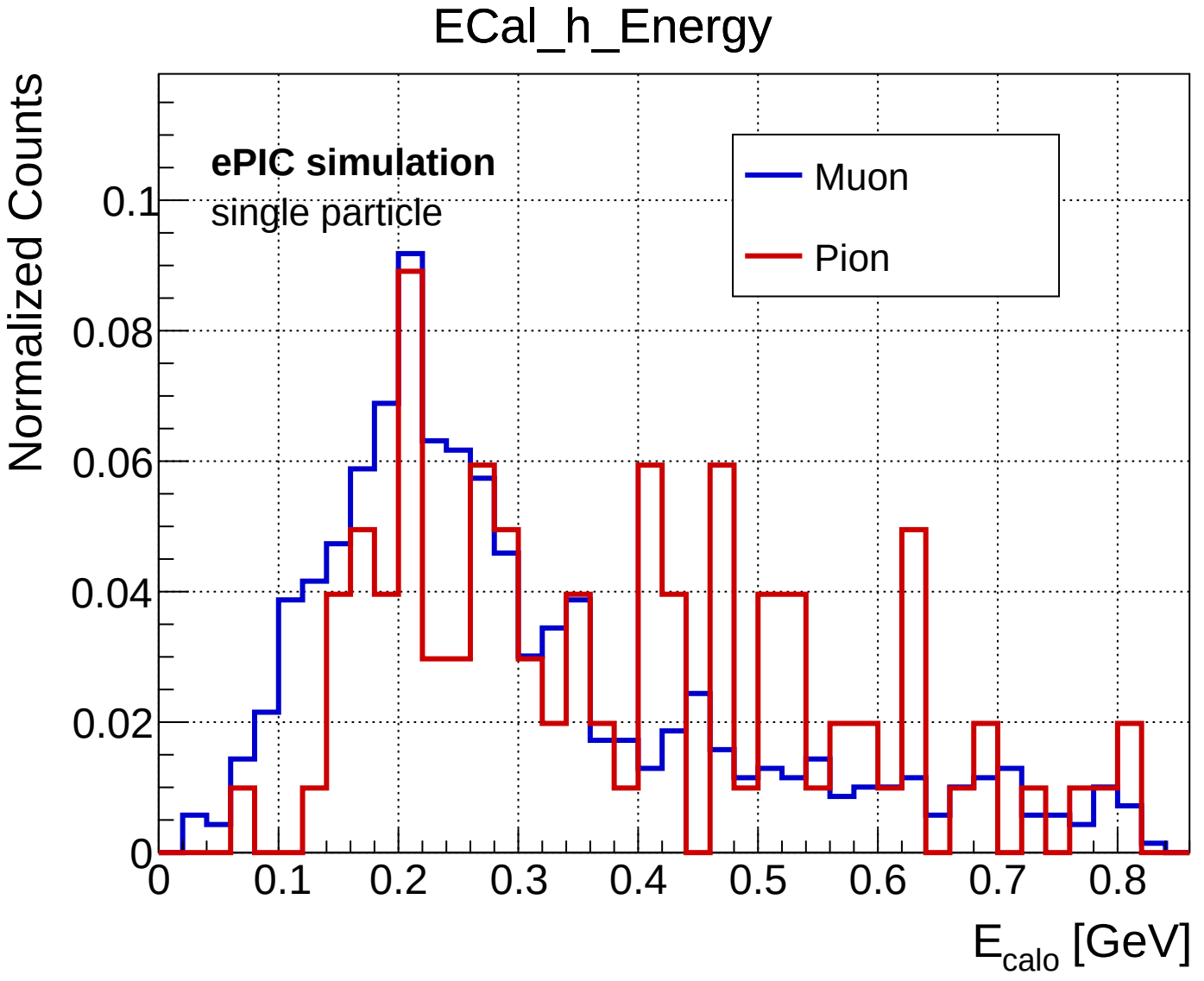

ECal_HitZ
Normalized Counts
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
ePIC simulation
single particle
Muon
Pion
−2500
−2000
−1500
−1000
−500
0
500
1000
Z [mm]

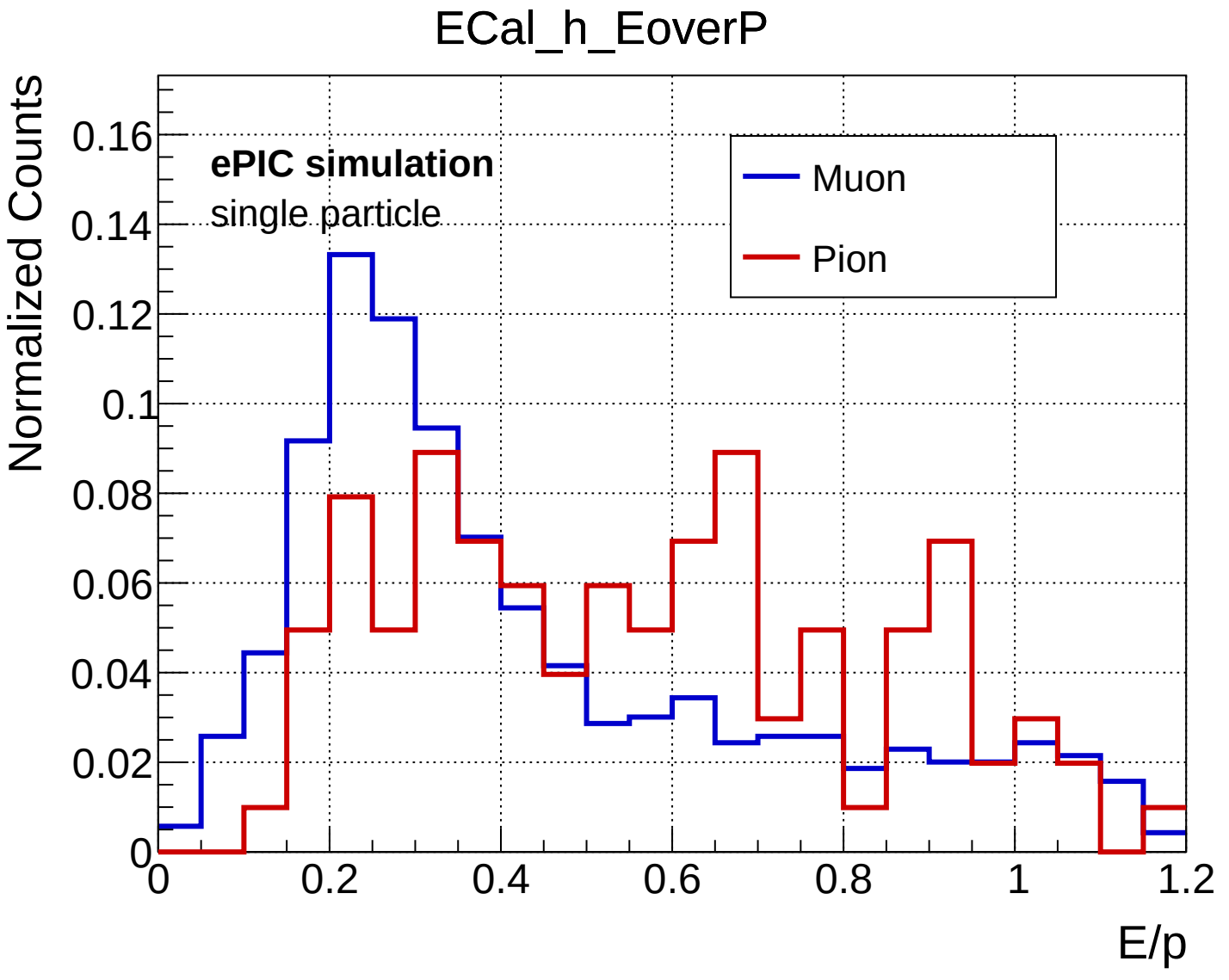

ECal_HitEnergy
ePIC simulation
single particle
Muon
Pion
Normalized Counts
Energy [GeV]
0.22
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18

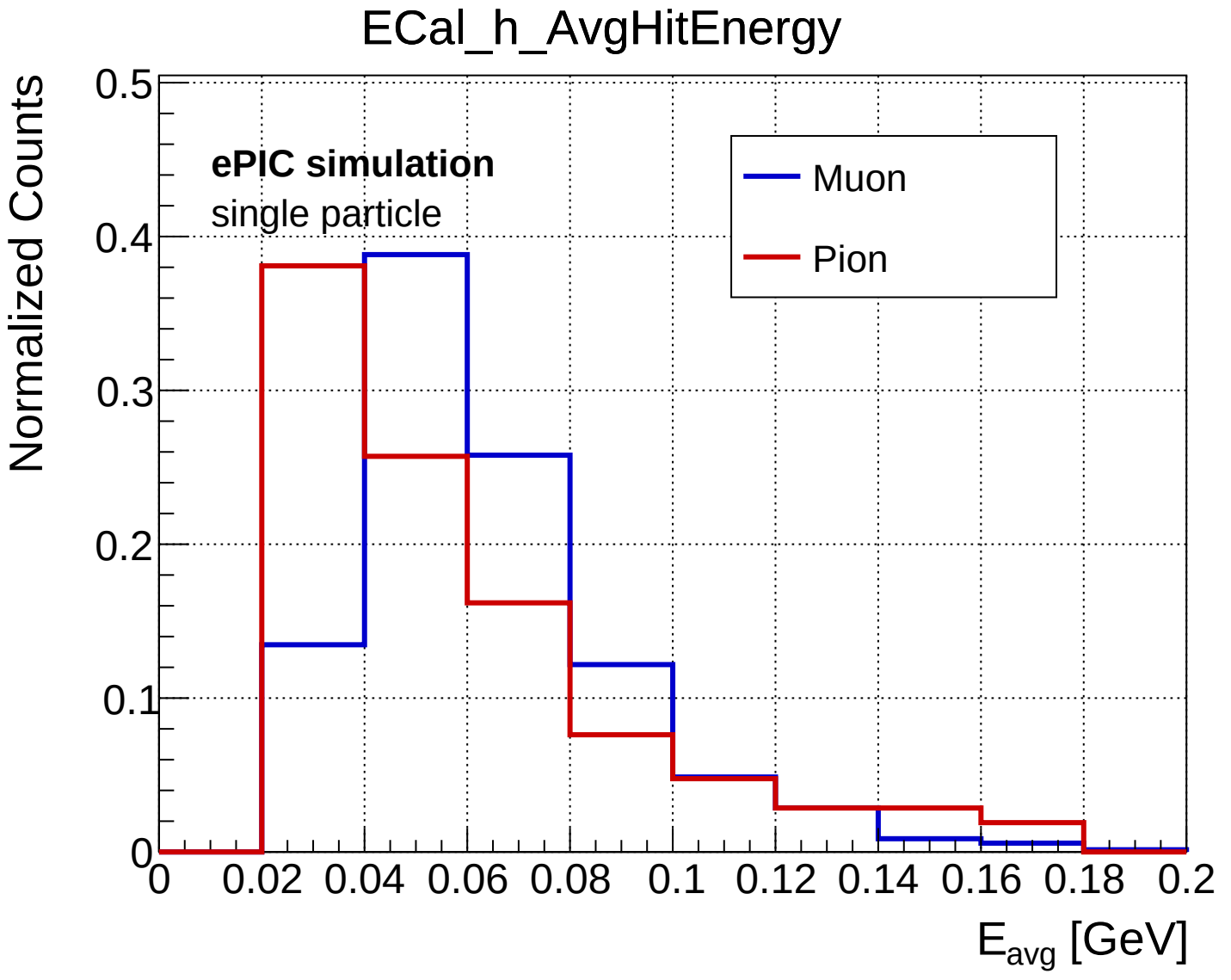

ECal_HitTime
ePIC simulation
single particle
Muon
Pion
Normalized Counts
0
0.05
0.1
0.15
0.2
0.25
0.3
2
4
6
8
10
12
14
16
Time [ns]

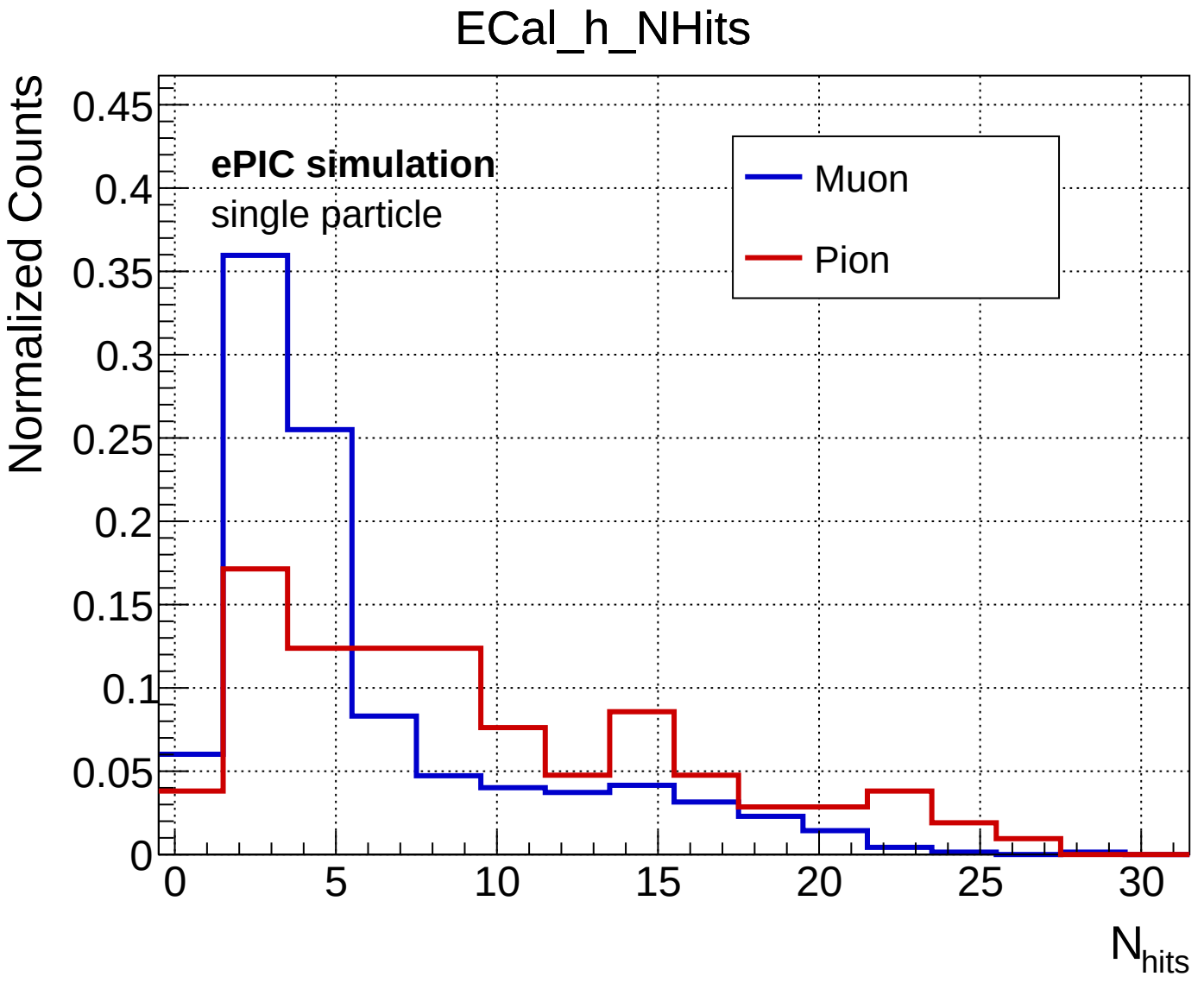

ECal_DeltaEta
Normalized Counts
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
ePIC simulation
single particle
Muon
Pion
–0.6
–0.4
–0.2
0
0.2
0.4
0.6
Δη

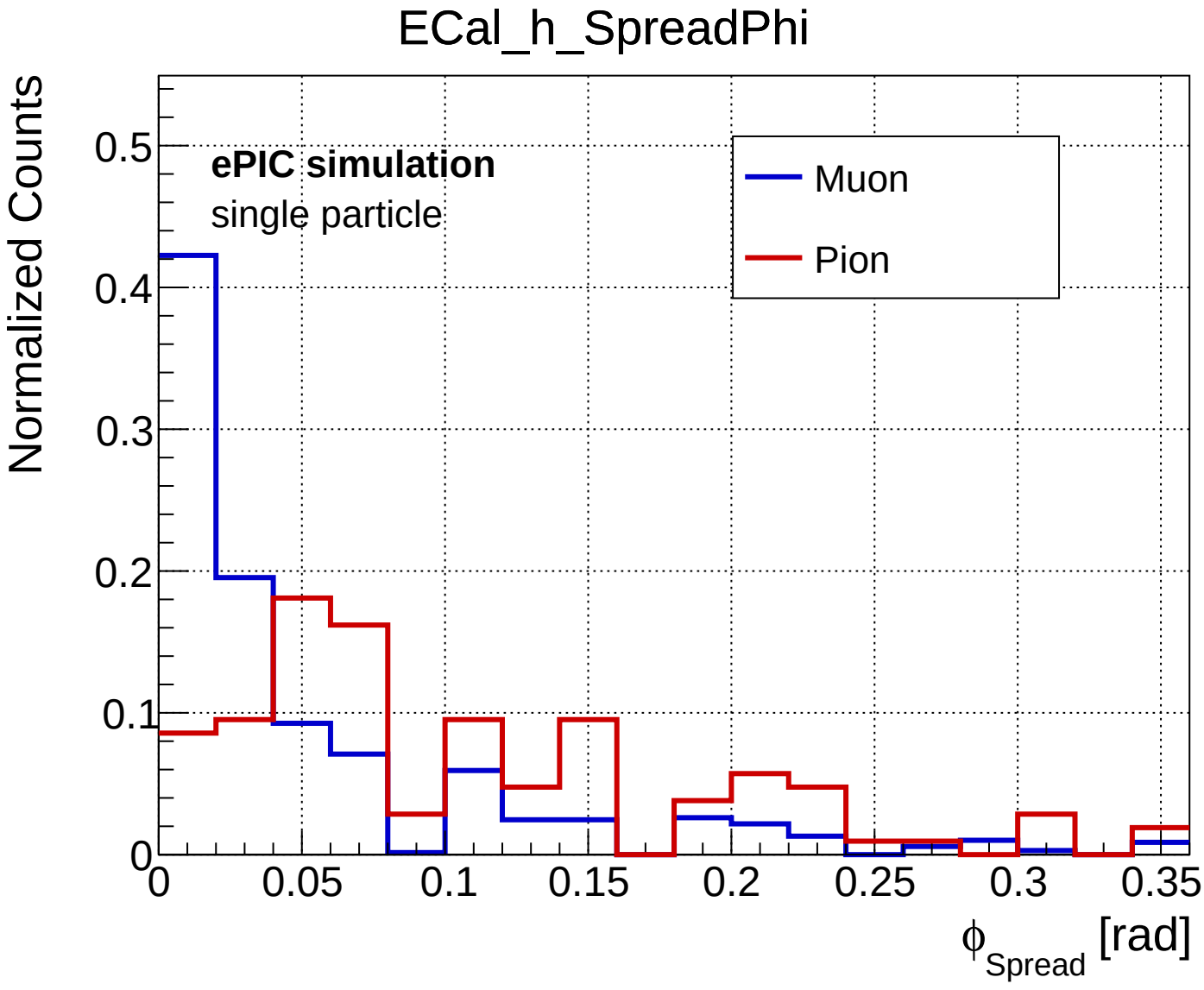

ECal_DeltaPhi
Normalized Counts
0.04
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
ePIC simulation
single particle
Muon
Pion
−0.8
−0.6
−0.4
−0.2
0
0.2
0.4
0.6
0.8
Δϕ

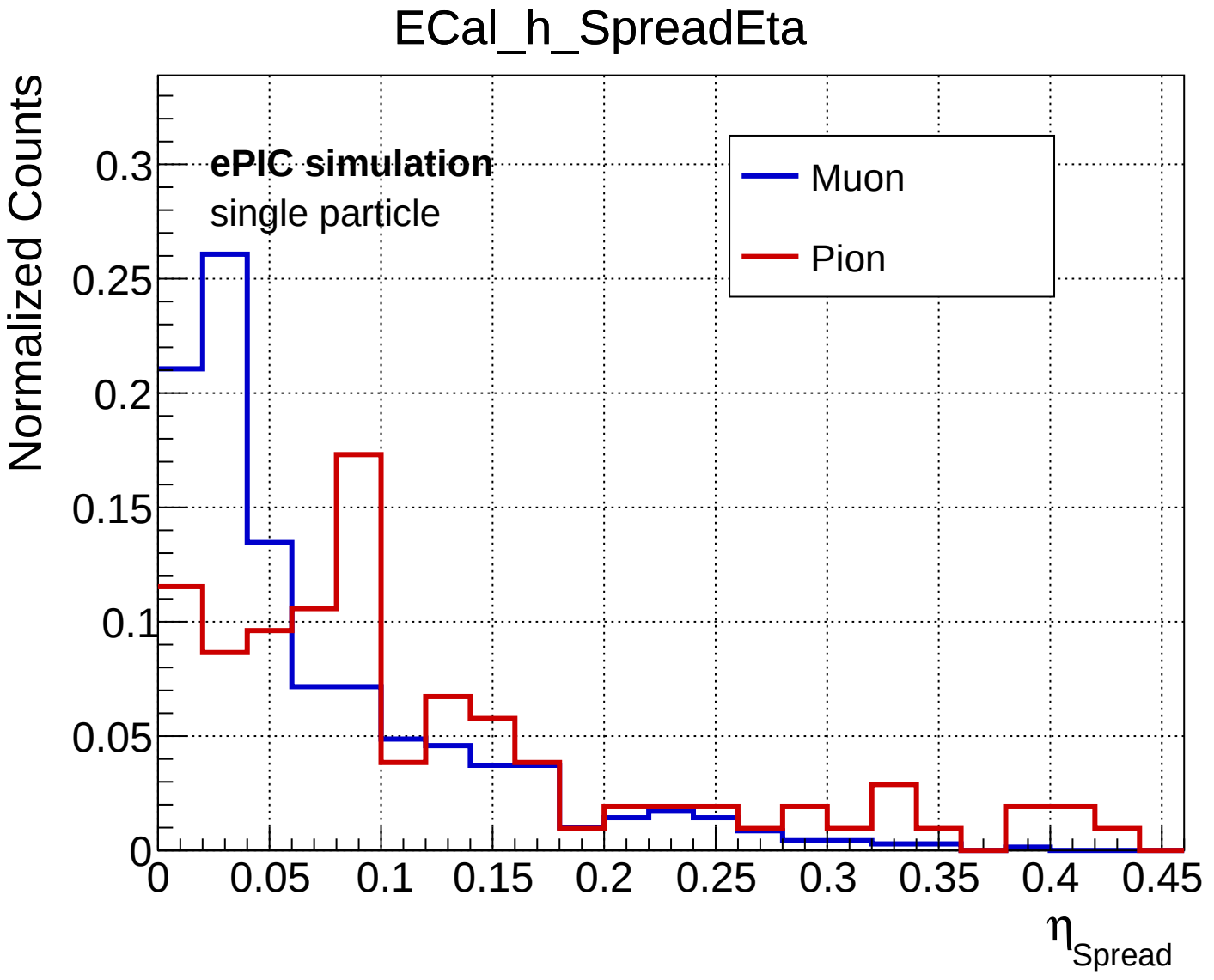

ECal_h_Energy
ePIC simulation
single particle
Muon
Pion
Normalized Counts
0
0.02
0.04
0.06
0.08
0.1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
E_calo [GeV]

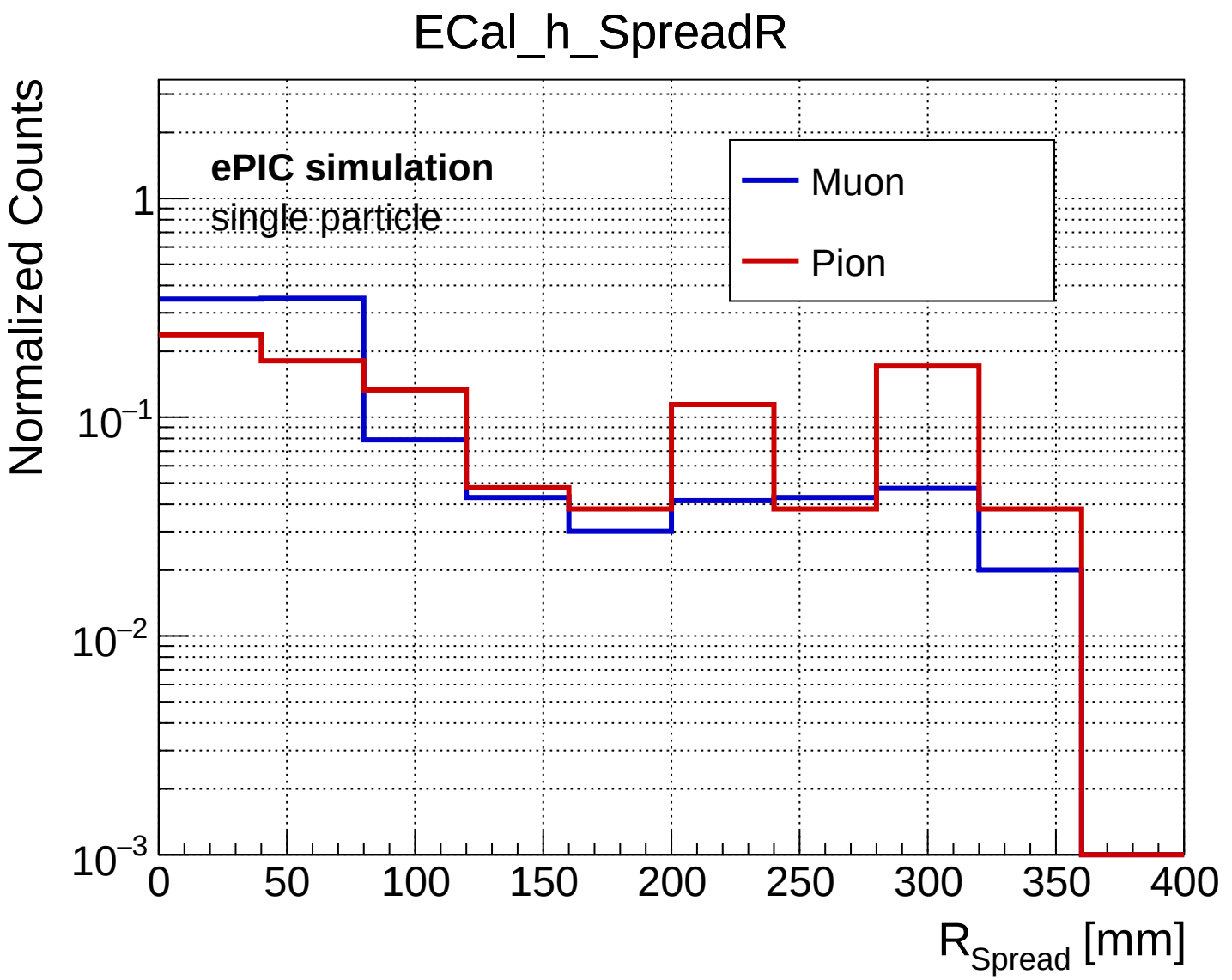

ECal_h_EoverP
Normalized Counts
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
ePIC simulation
single particle
Muon
Pion
0
0.2
0.4
0.6
0.8
1
1.2
E/p

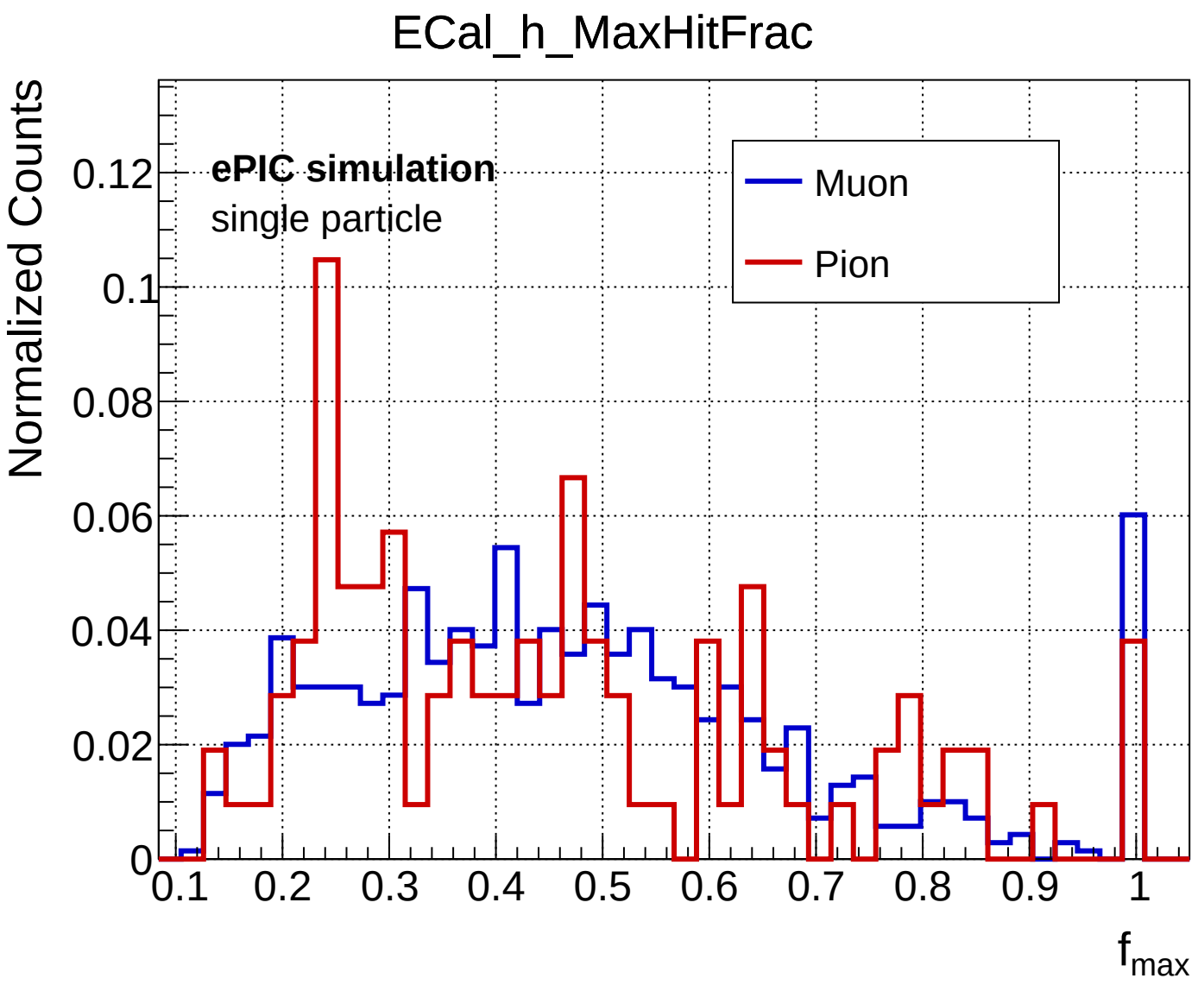

ECal_h_AvgHitEnergy
Normalized Counts
0.5
0.4
0.3
0.2
0.1
0
ePIC simulation
single particle
Muon
Pion
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
E_avg [GeV]

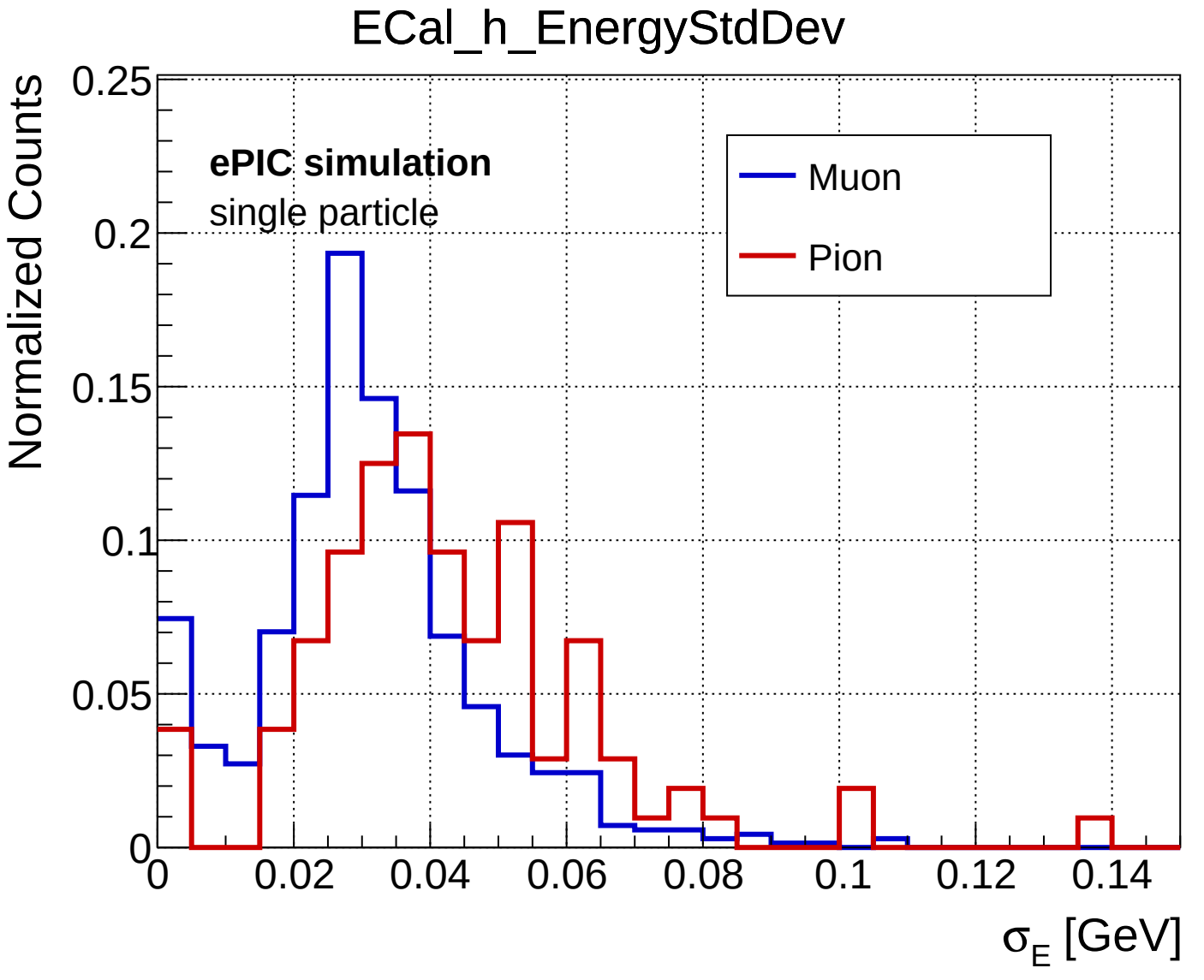

ECal_h_NHits
Normalized Counts
0.45
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
0
5
10
15
20
25
30
N hits
ePIC simulation
single particle
Muon
Pion

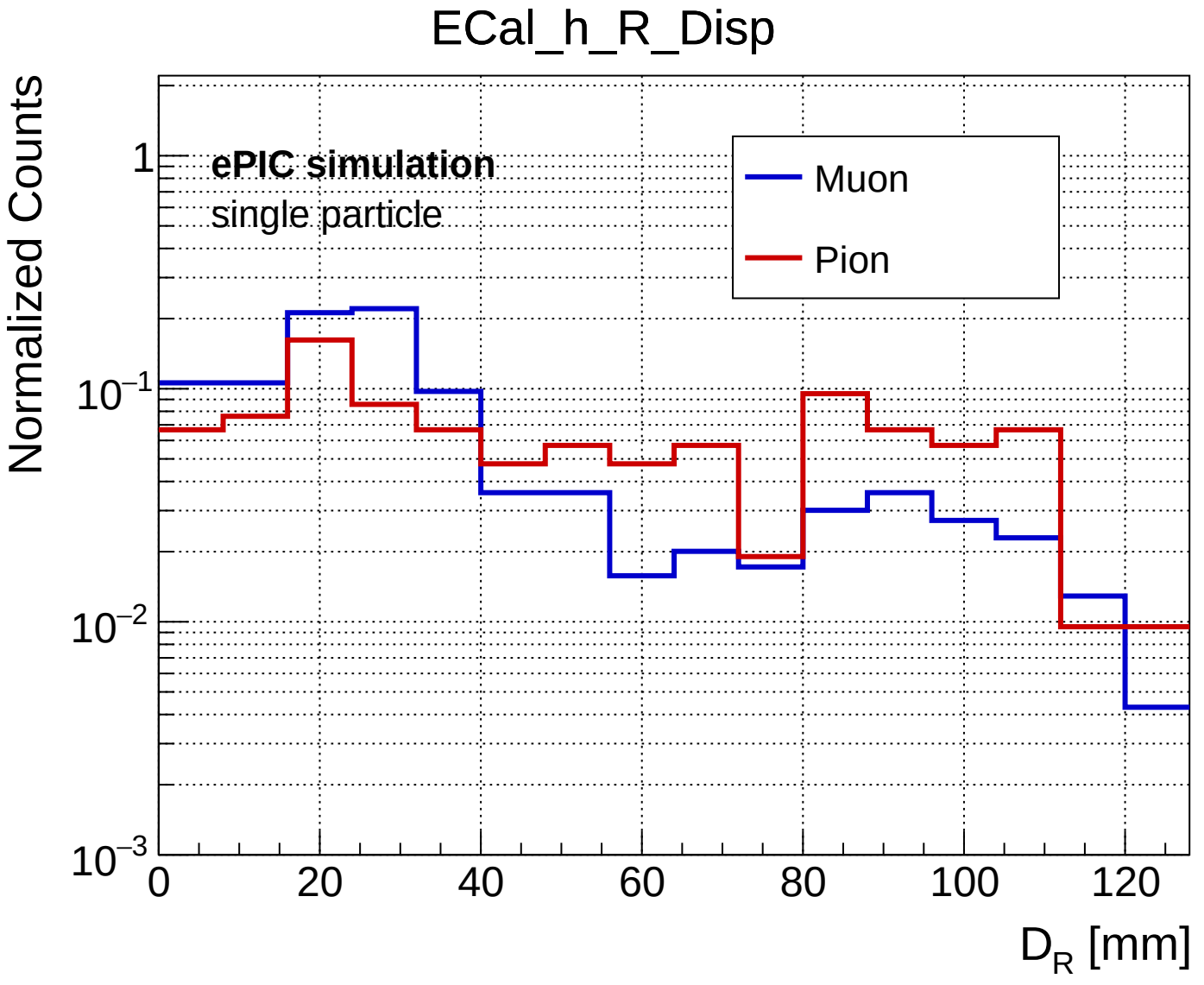

ECal_h_SpreadPhi
Normalized Counts
ePIC simulation
single particle
Muon
Pion
0.5
0.4
0.3
0.2
0.1
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
ϕ_{Spread} [rad]

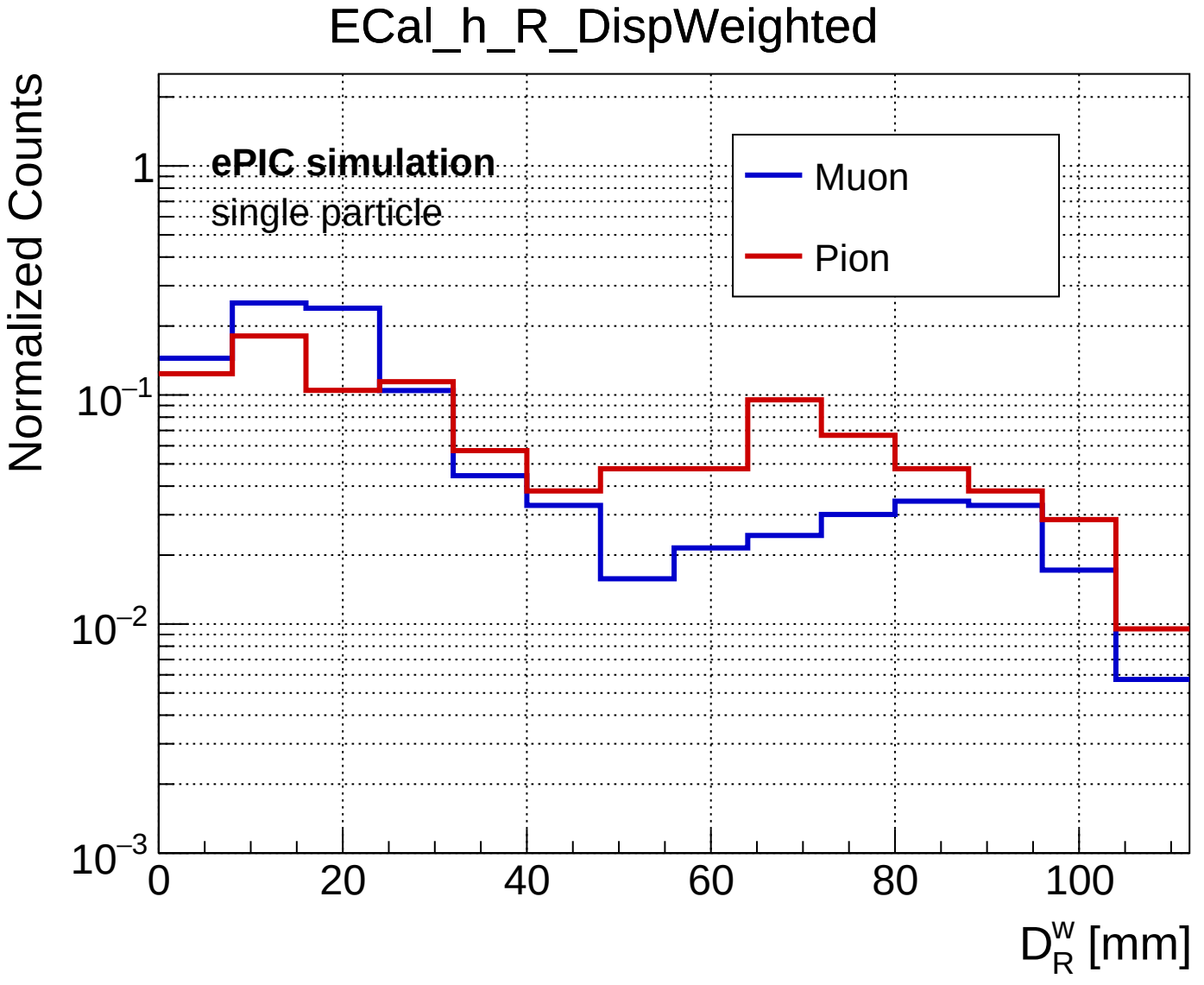

ECal_h_SpreadEta
Normalized Counts
ePIC simulation
single particle
Muon
Pion
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
η Spread

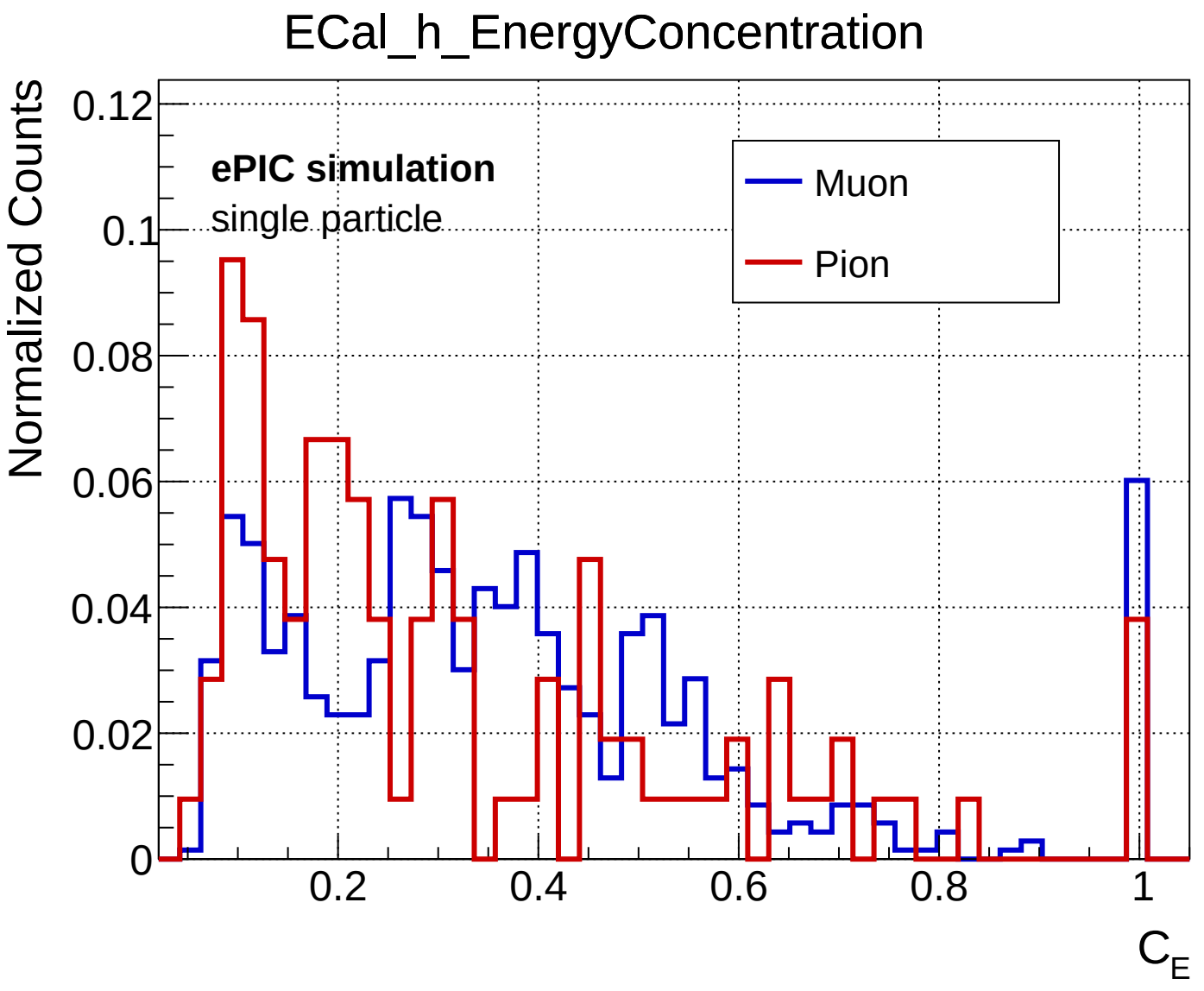

ECal_h_SpreadR
ePIC simulation
single particle
Muon
Pion
Normalized Counts
1
10^{-1}
10^{-2}
10^{-3}
0
50
100
150
200
250
300
350
400
R_{Spread} [mm]

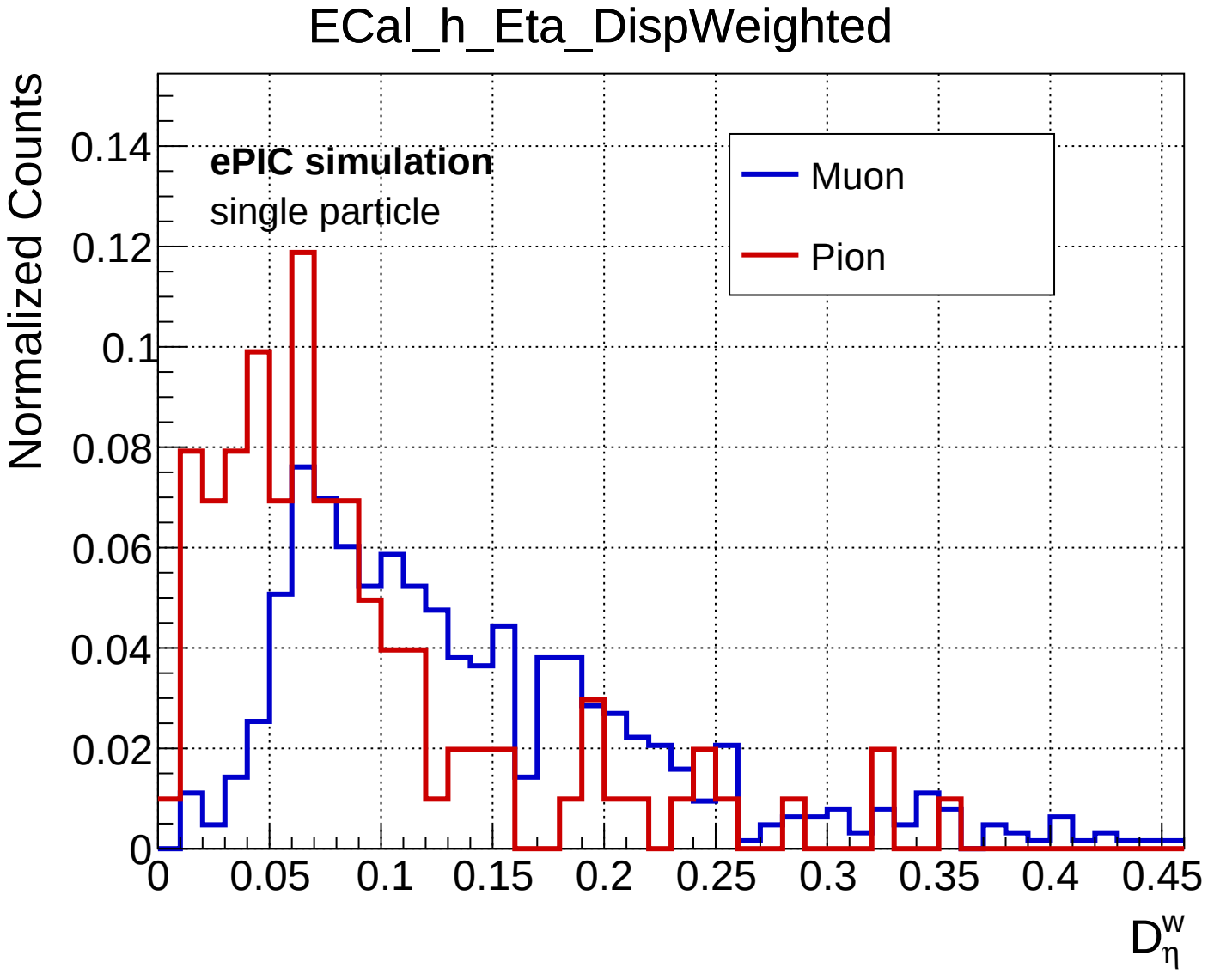

ECal_h_MaxHitFrac
Normalized Counts
0.12
0.1
0.08
0.06
0.04
0.02
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
f_{max}
ePIC simulation
single particle
Muon
Pion

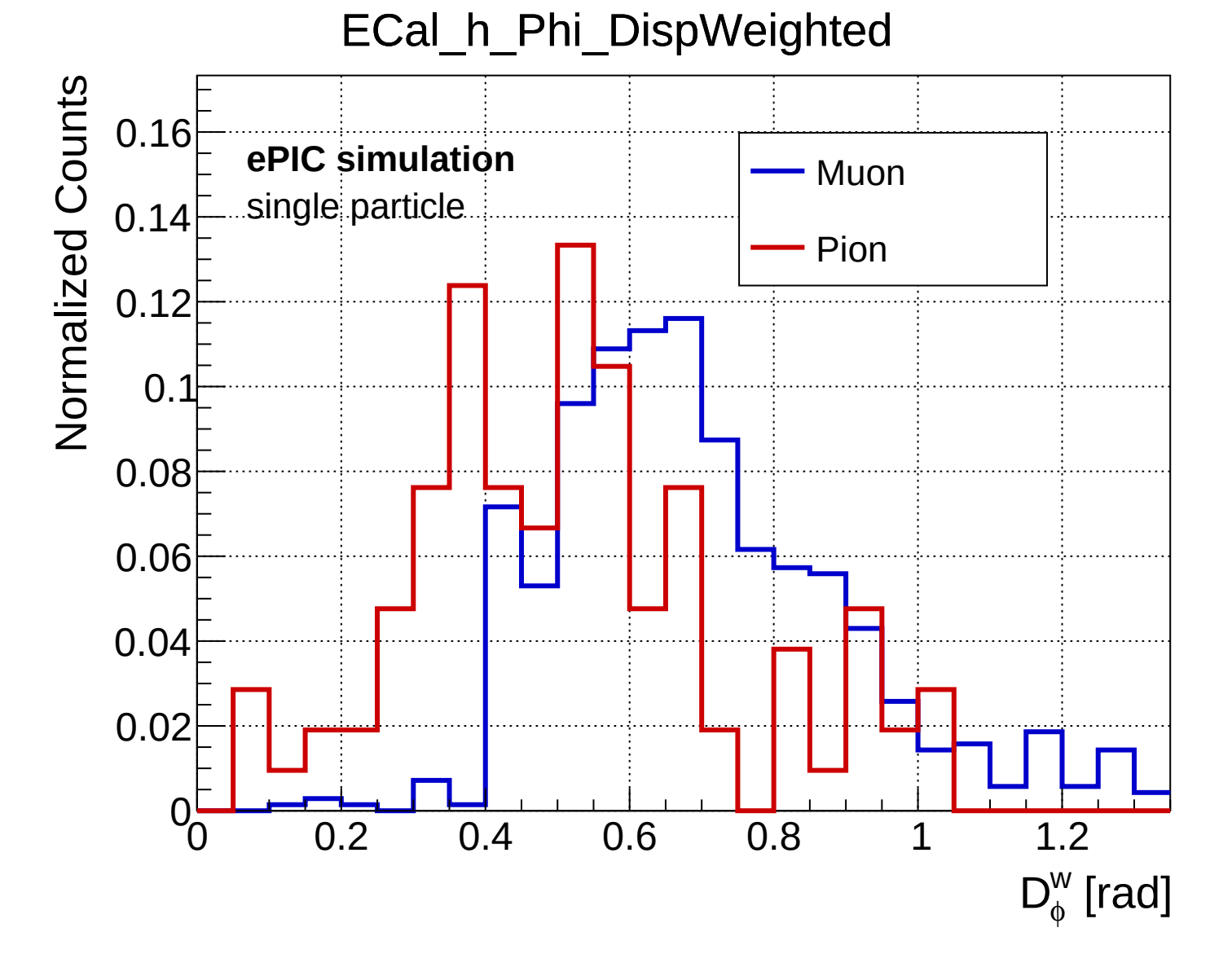

ECal_h_EnergyStdDev
ePIC simulation
single particle
Muon
Pion
Normalized Counts
0.25
0.2
0.15
0.1
0.05
0
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
σE [GeV]

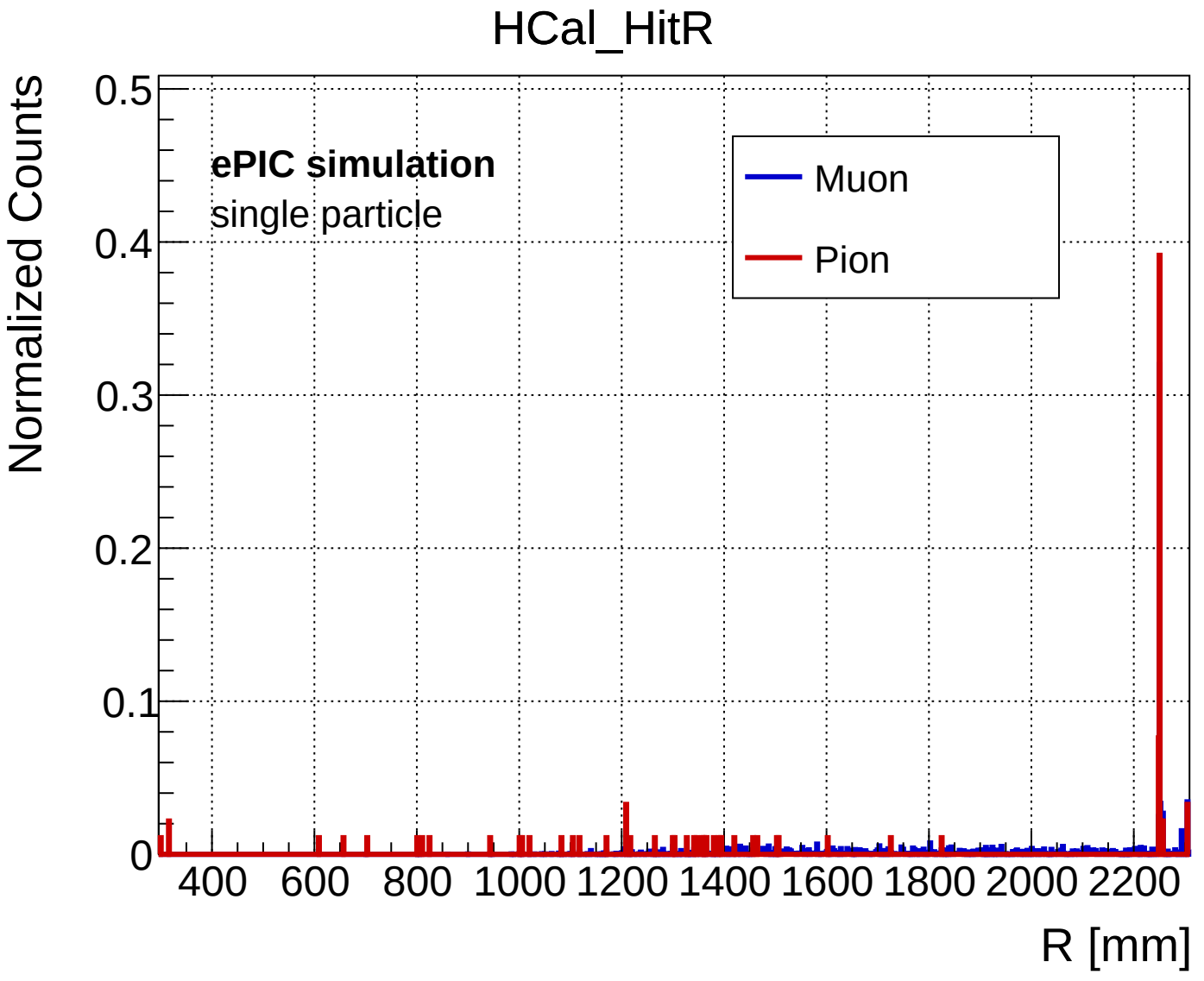

ECal_h_R_Disp
ePIC simulation
single particle
Muon
Pion
Normalized Counts
1
10^{-1}
10^{-2}
10^{-3}
0
20
40
60
80
100
120
D_R [mm]

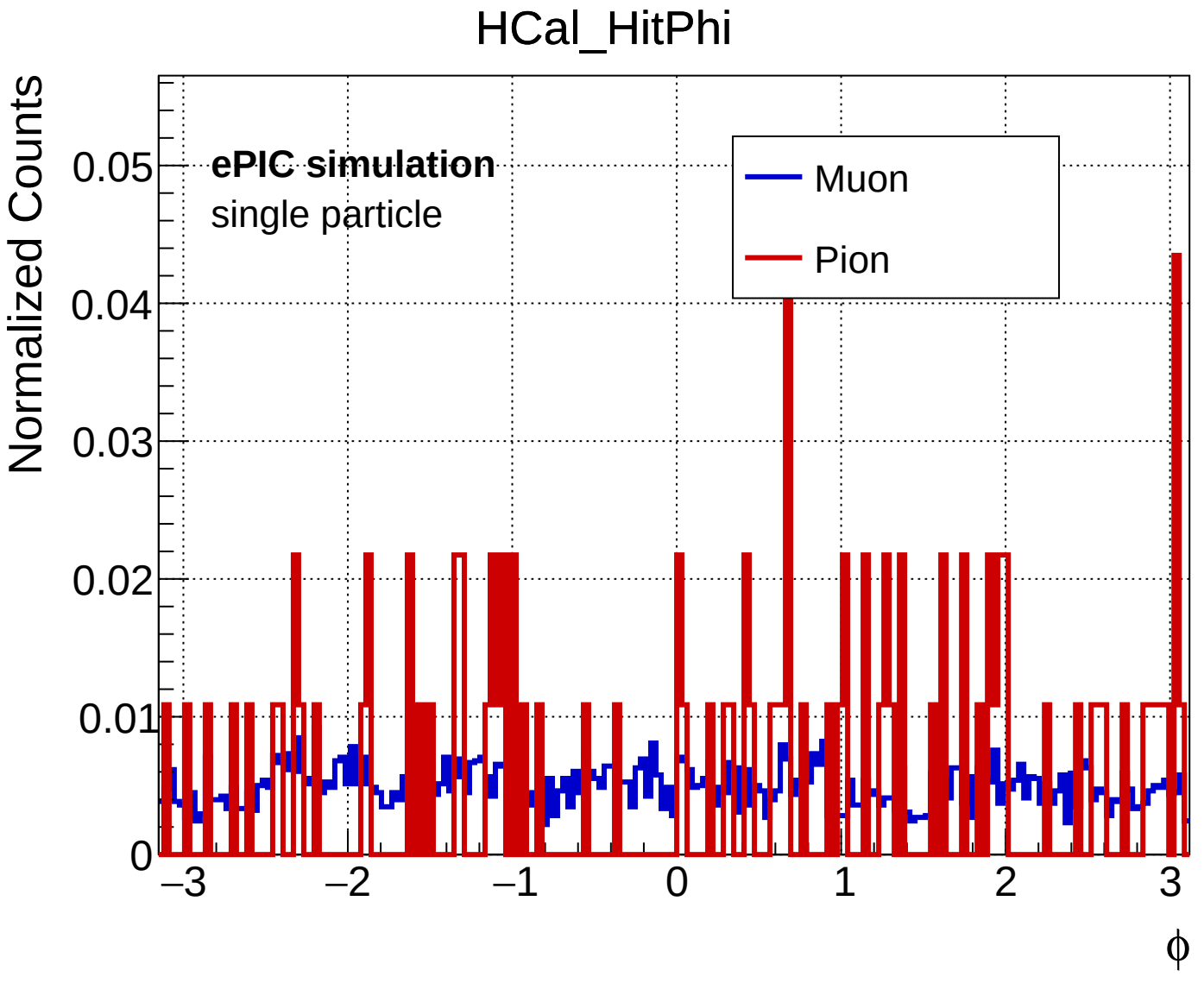

ECal_h_R_DispWeighted
Normalized Counts
ePIC simulation
single particle
Muon
Pion
1
10^{-1}
10^{-2}
10^{-3}
0
20
40
60
80
100
D_R^W [mm]

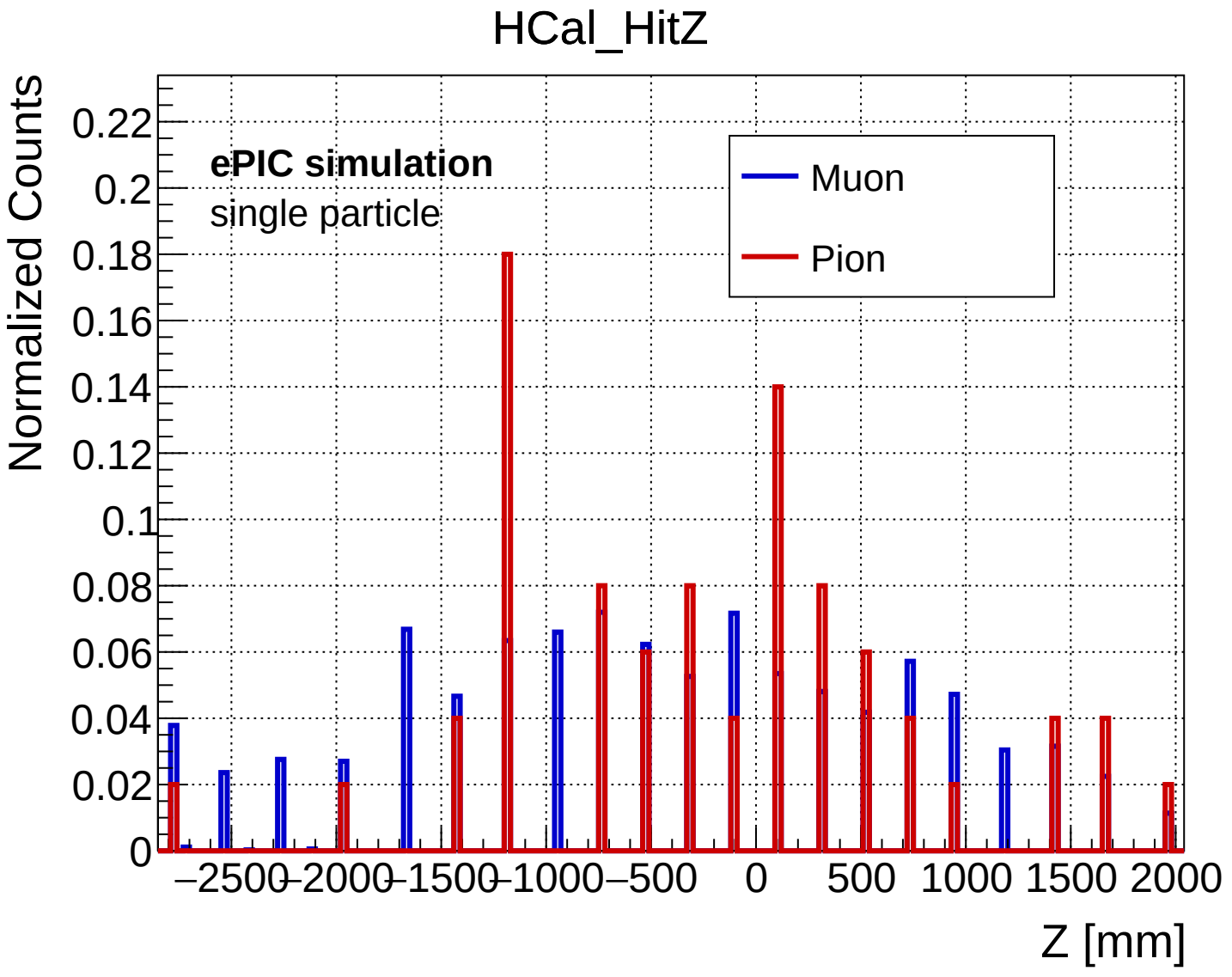

ECal_h_EnergyConcentration
Normalized Counts
0.12
0.1
0.08
0.06
0.04
0.02
0
0.2
0.4
0.6
0.8
1
C_E
ePIC simulation
single particle
Muon
Pion

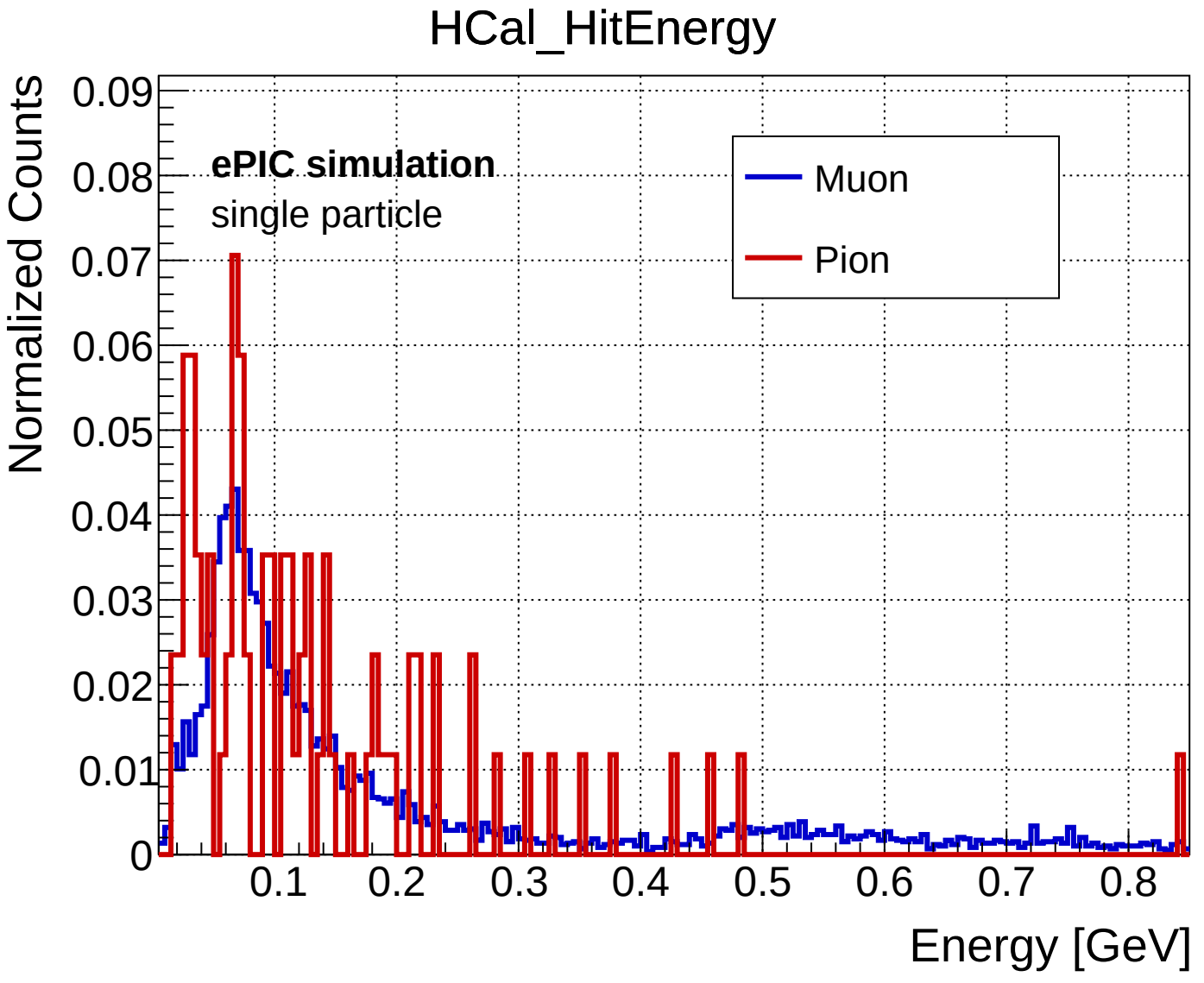

ECal_h_Eta_DispWeighted
Normalized Counts
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
D_η^w
ePIC simulation
single particle
Muon
Pion

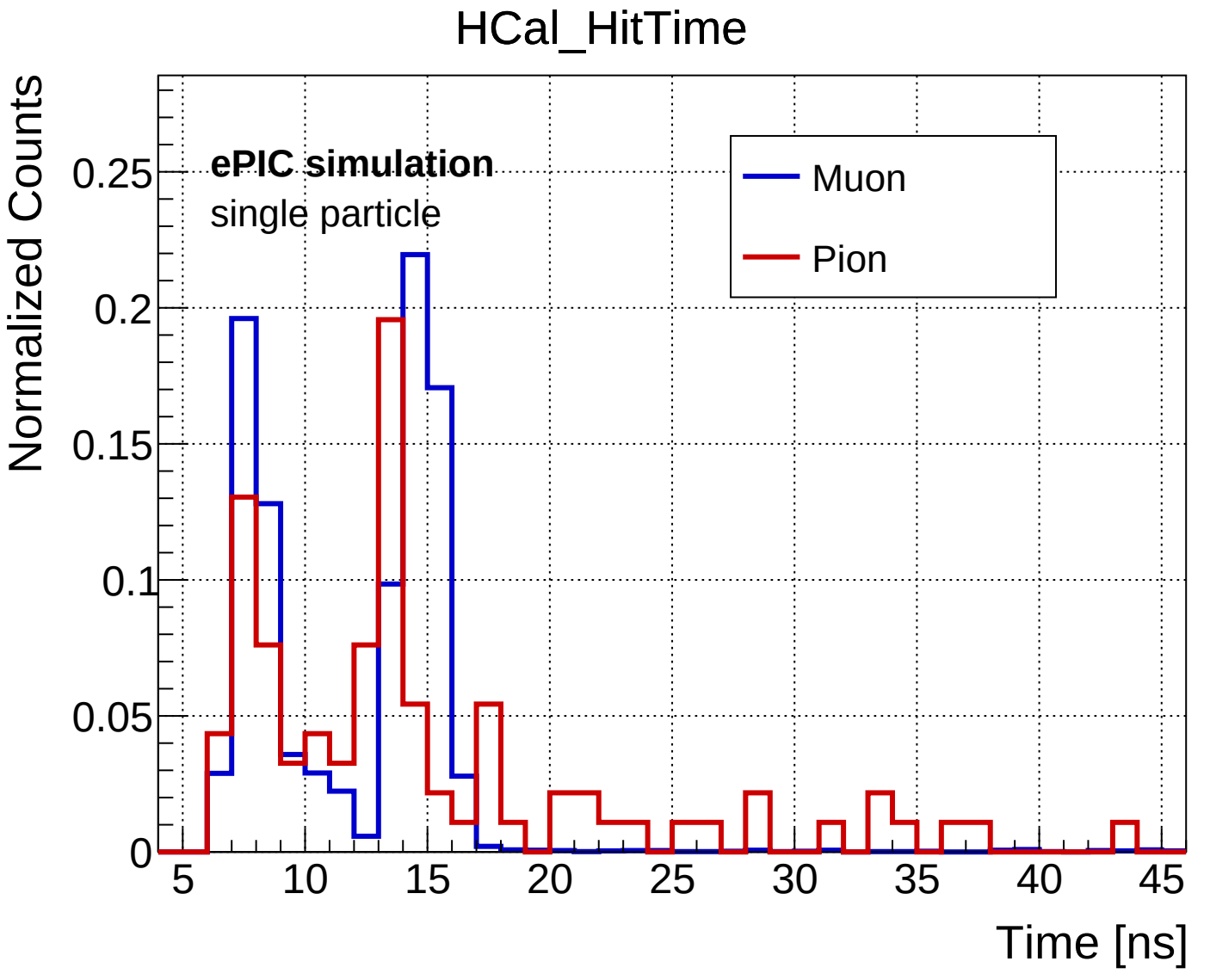

ECal_h_Phi_DispWeighted
Normalized Counts
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
ePIC simulation
single particle
Muon
Pion
0
0.2
0.4
0.6
0.8
1
1.2
D_{ϕ}^{w} [rad]

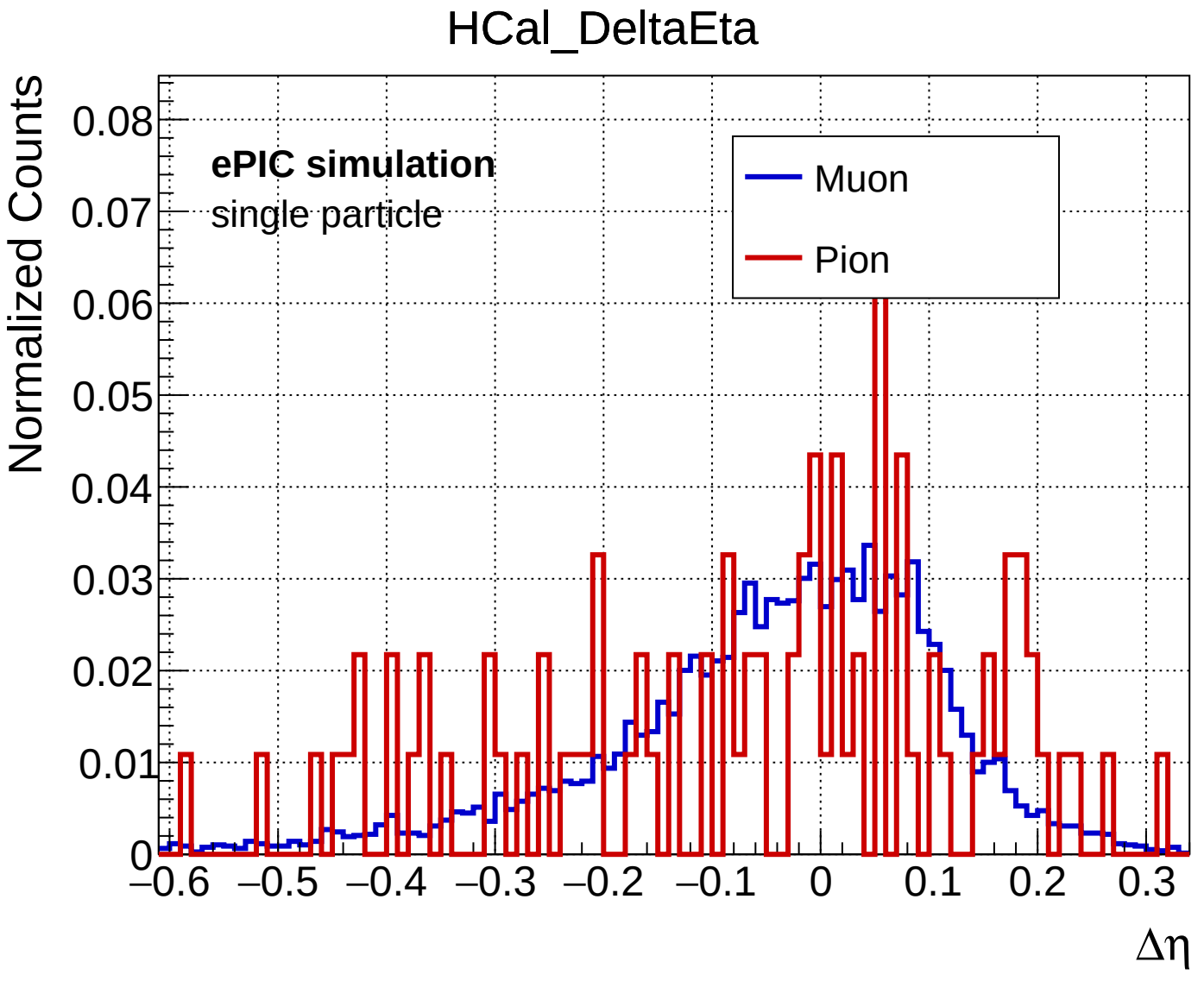

HCal_HitR
Normalized Counts
0.5
0.4
0.3
0.2
0.1
0
ePIC simulation
single particle
Muon
Pion
400
600
800
1000
1200
1400
1600
1800
2000
2200
R [mm]

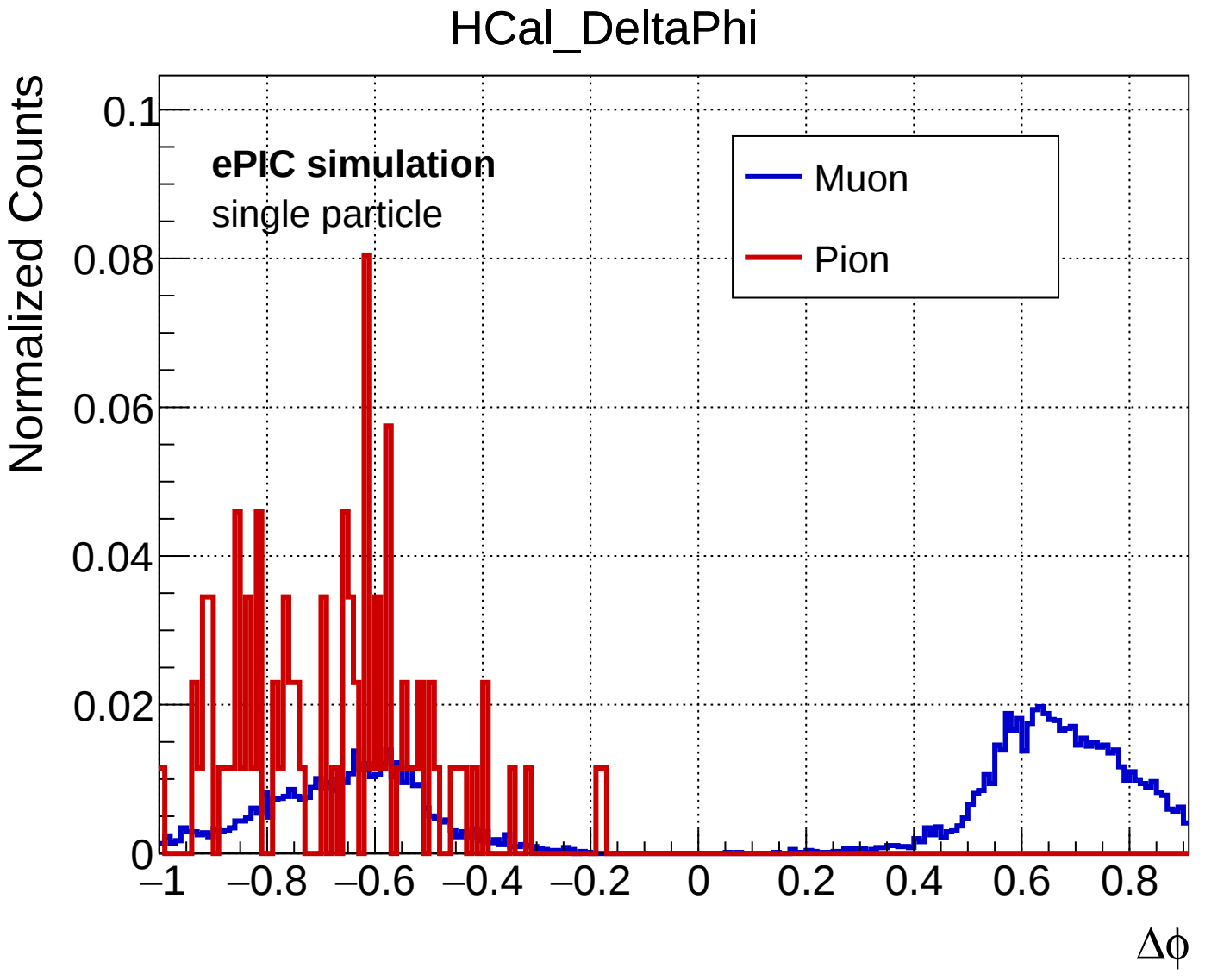

HCal_HitPhi
Normalized Counts
0.05
0.04
0.03
0.02
0.01
0
ePIC simulation
single particle
Muon
Pion
−3
−2
−1
0
1
2
3
ϕ

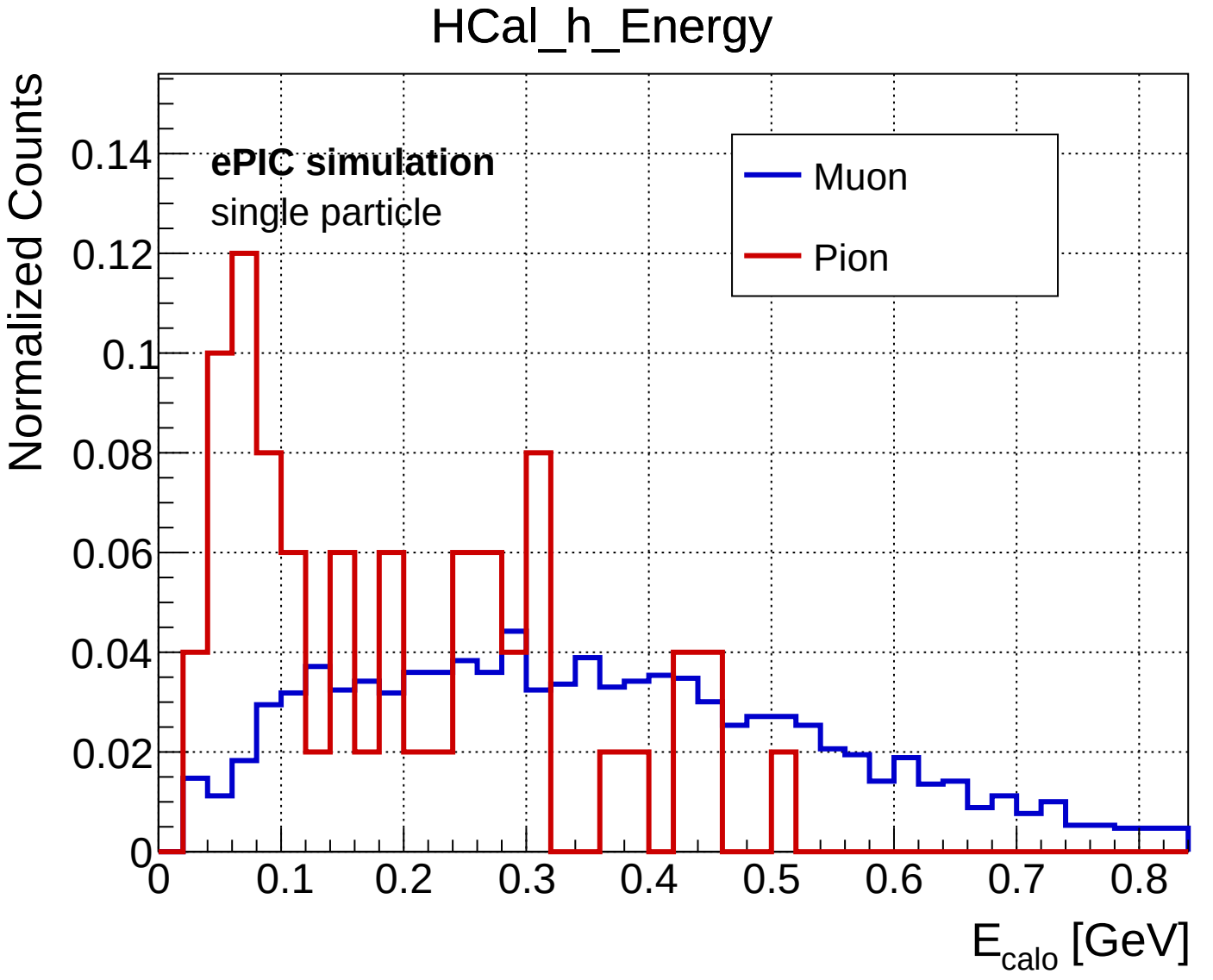

HCal_HitZ
Normalized Counts
0.22
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
ePIC simulation
single particle
Muon
Pion
−2500
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Z [mm]

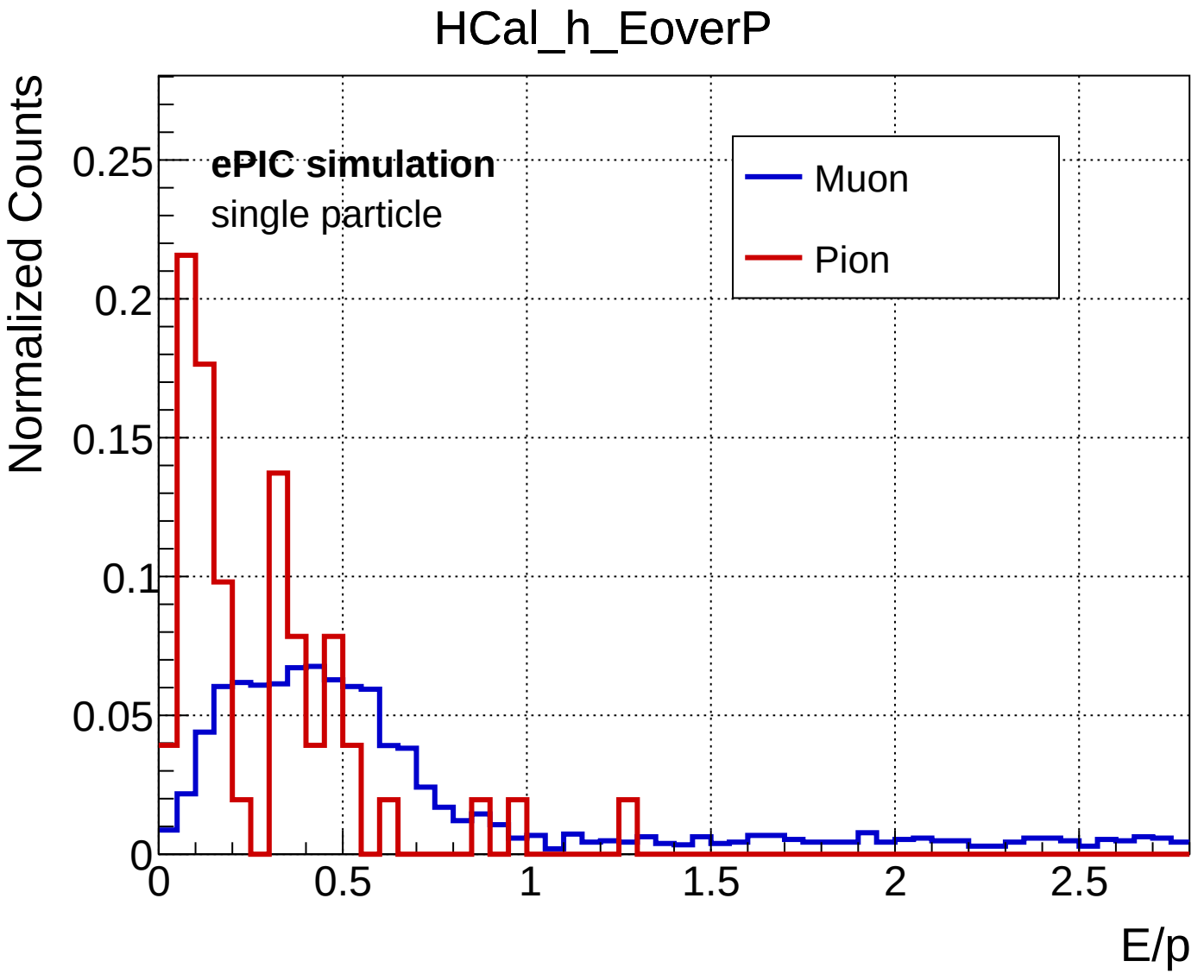

HCal_HitEnergy
Normalized Counts
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
ePIC simulation
single particle
Muon
Pion
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Energy [GeV]

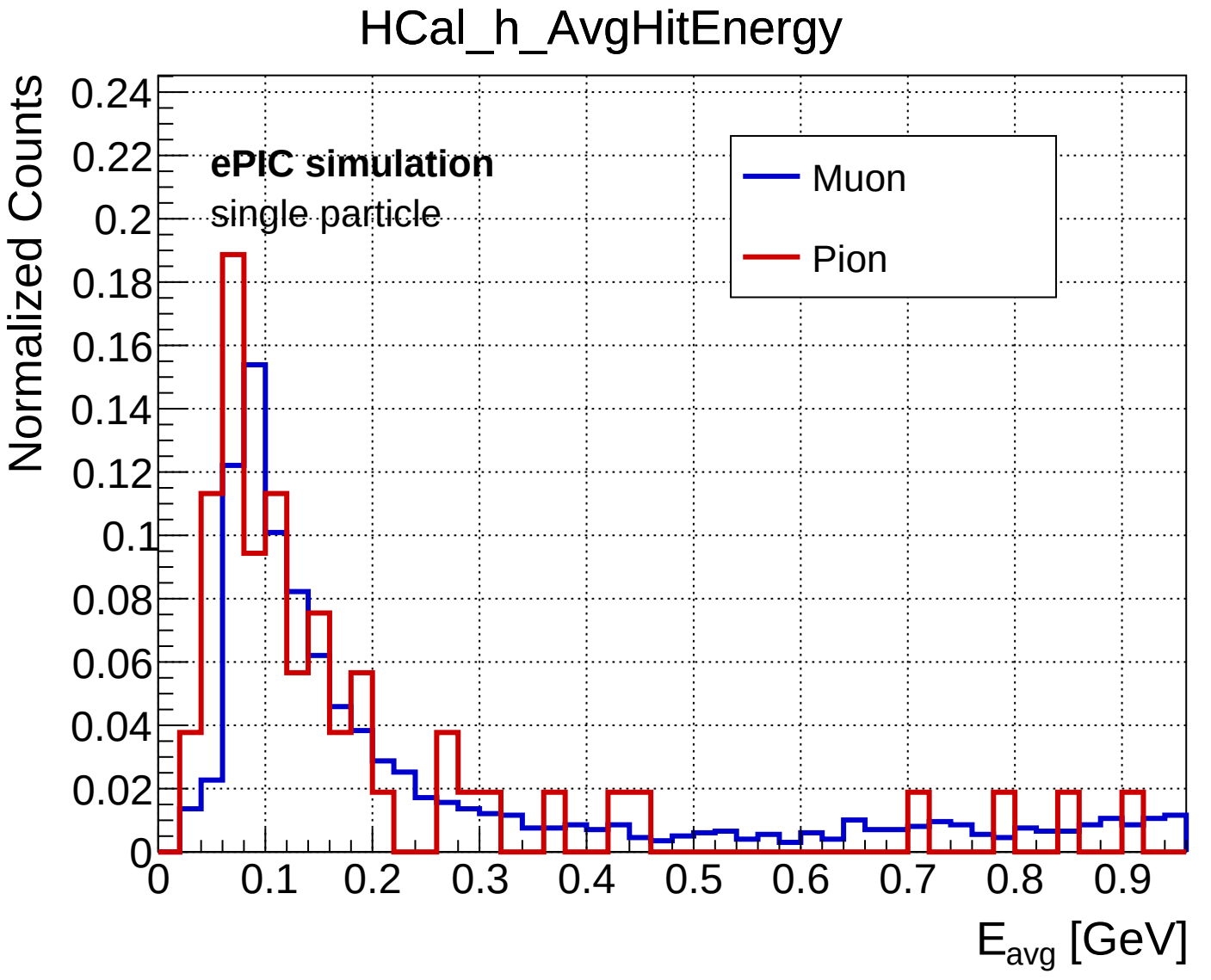

HCal_HitTime
ePIC simulation
single particle
Muon
Pion
Normalized Counts
0.25
0.2
0.15
0.1
0.05
0
5
10
15
20
25
30
35
40
45
Time [ns]

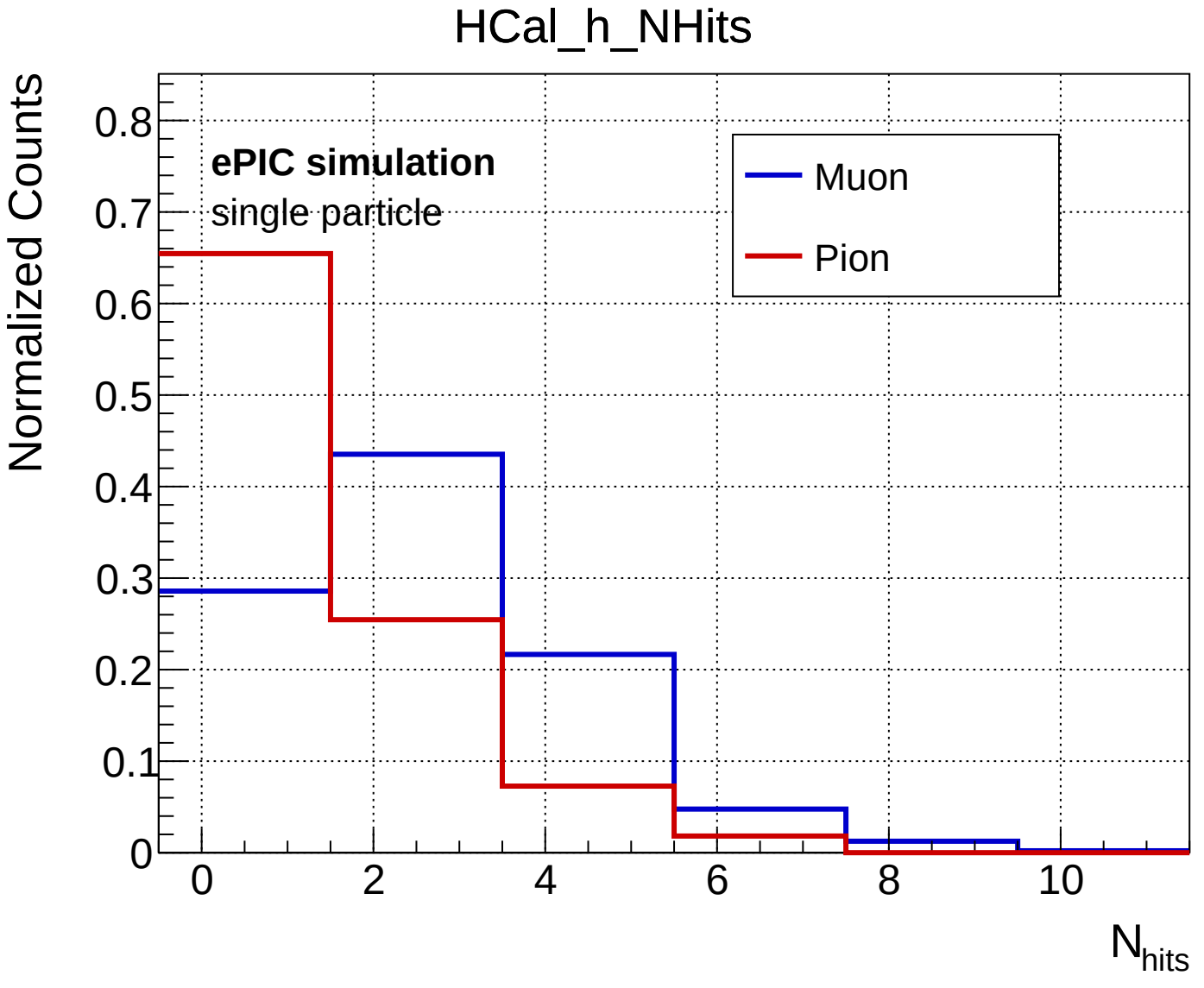

HCal_DeltaEta
Normalized Counts
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
ePIC simulation
single particle
Muon
Pion
–0.6
–0.5
–0.4
–0.3
–0.2
–0.1
0
0.1
0.2
0.3
Δη

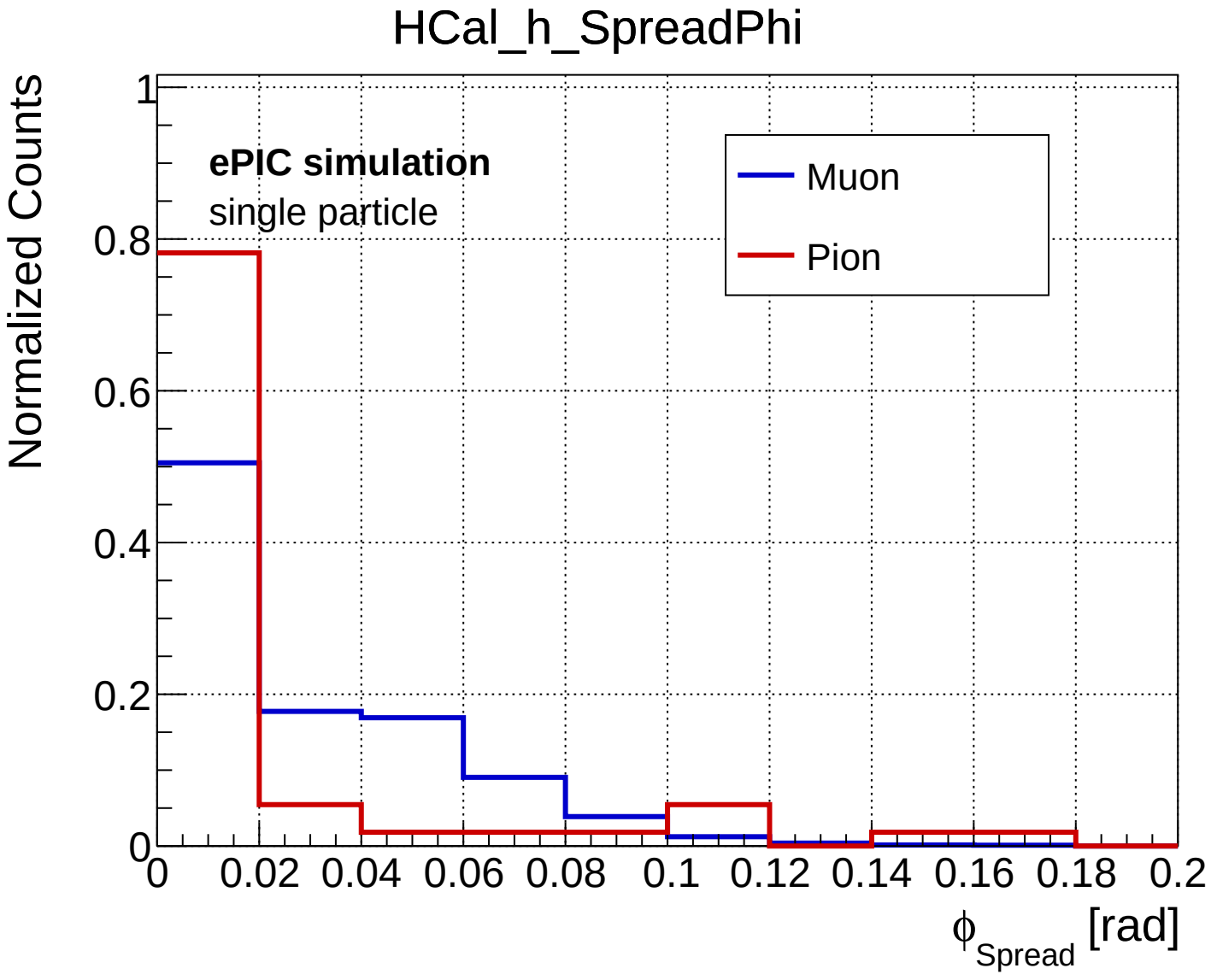

HCal_DeltaPhi
Normalized Counts
0.1
0.08
0.06
0.04
0.02
0
ePIC simulation
single particle
Muon
Pion
–1
–0.8
–0.6
–0.4
–0.2
0
0.2
0.4
0.6
0.8
Δϕ

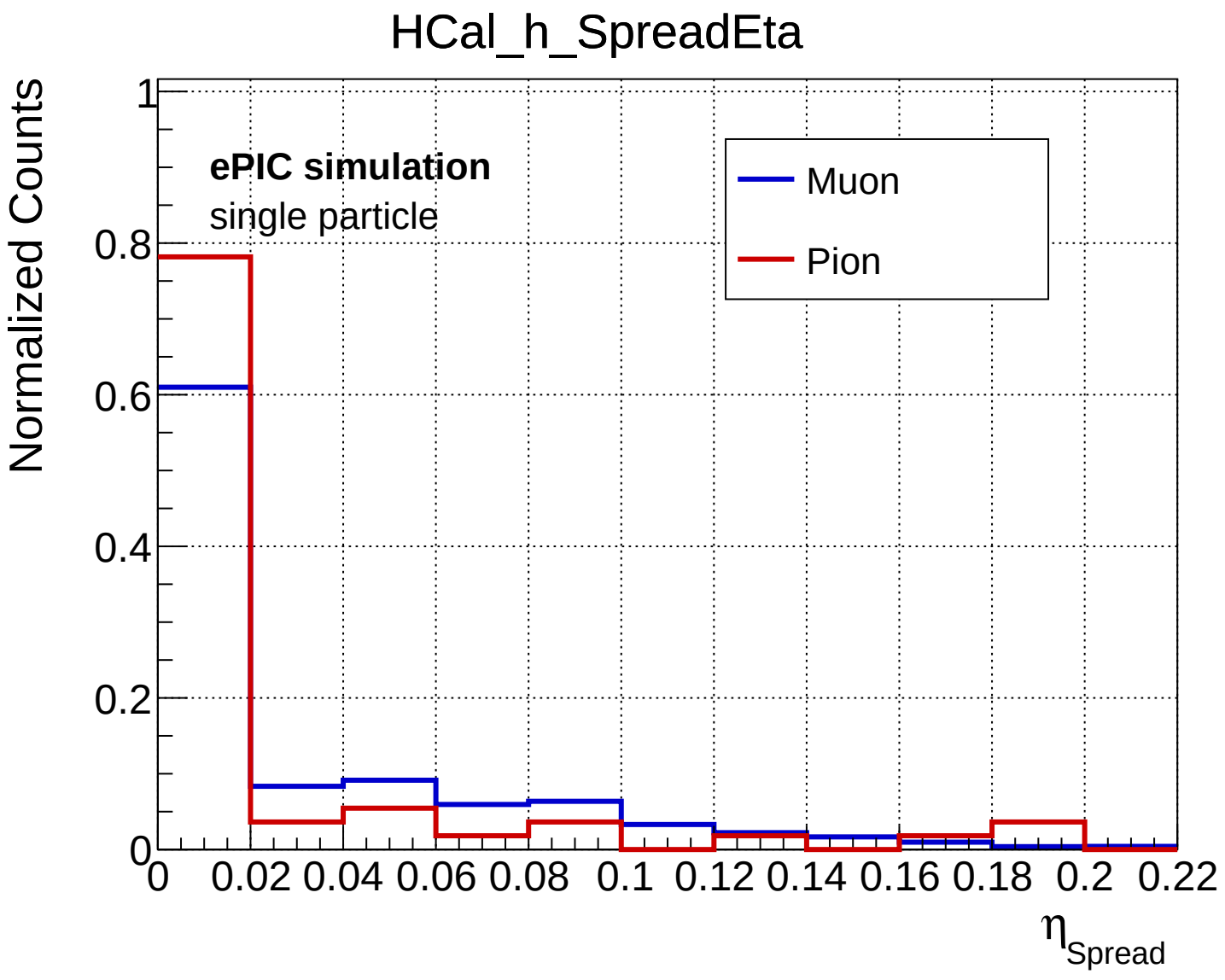

HCal_h_Energy
Normalized Counts
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
ePIC simulation
single particle
Muon
Pion
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
E_calo [GeV]

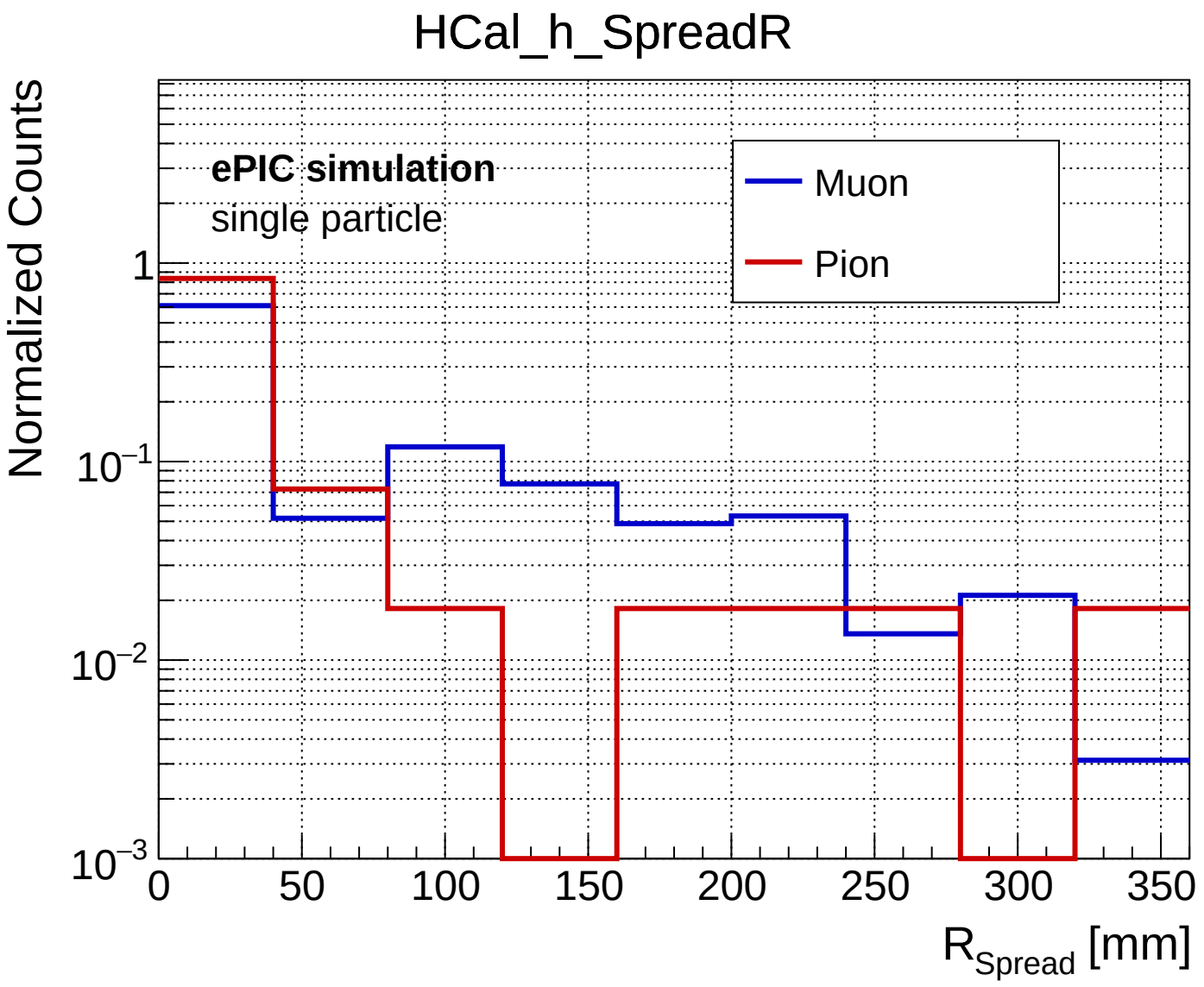

HCal_h_EoverP
Normalized Counts
0.25
0.2
0.15
0.1
0.05
0
0
0.5
1
1.5
2
2.5
E/p
ePIC simulation
single particle
Muon
Pion

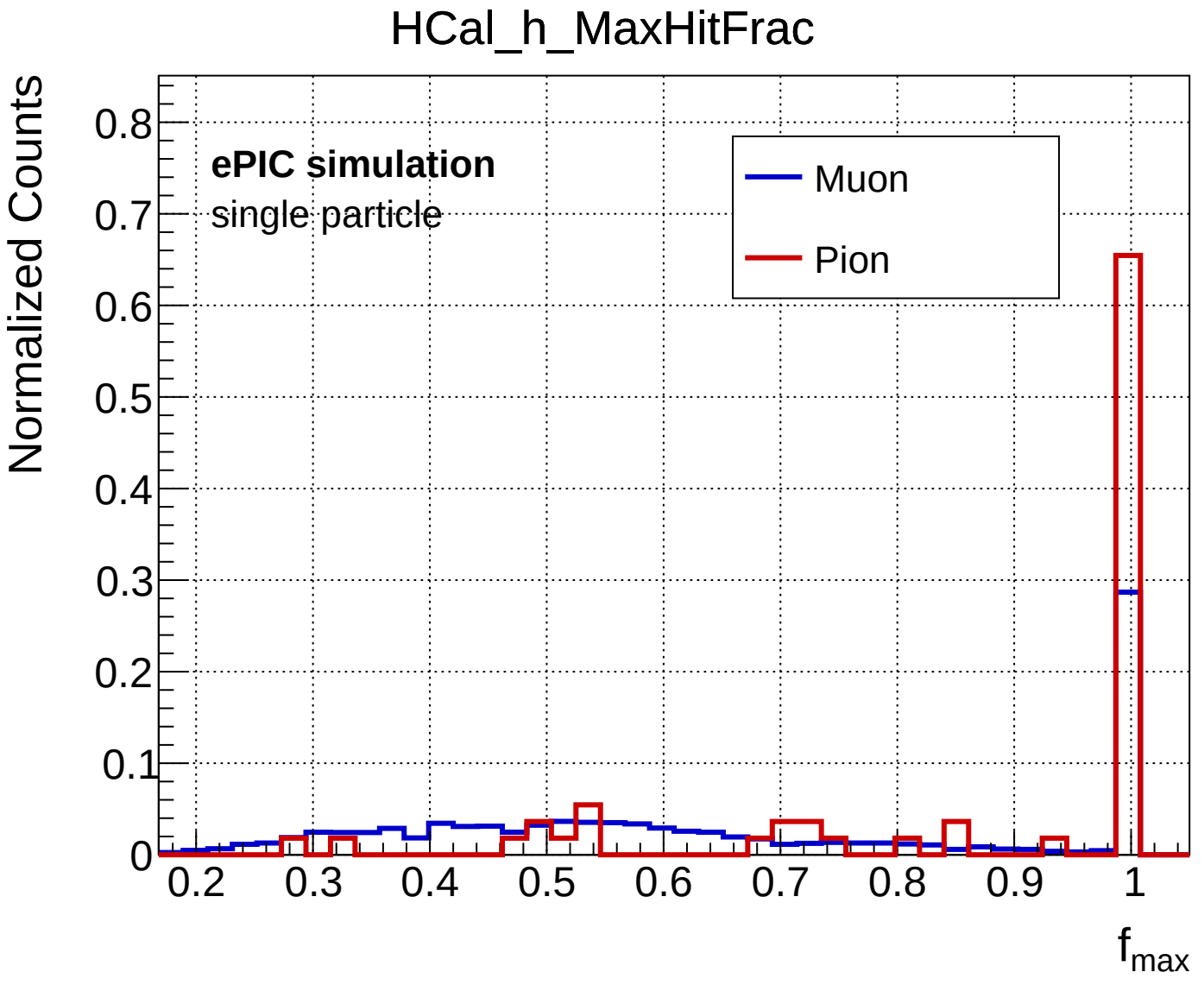

HCal_h_AvgHitEnergy
Normalized Counts
0.24
0.22
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
ePIC simulation
single particle
Muon
Pion
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
E_avg [GeV]

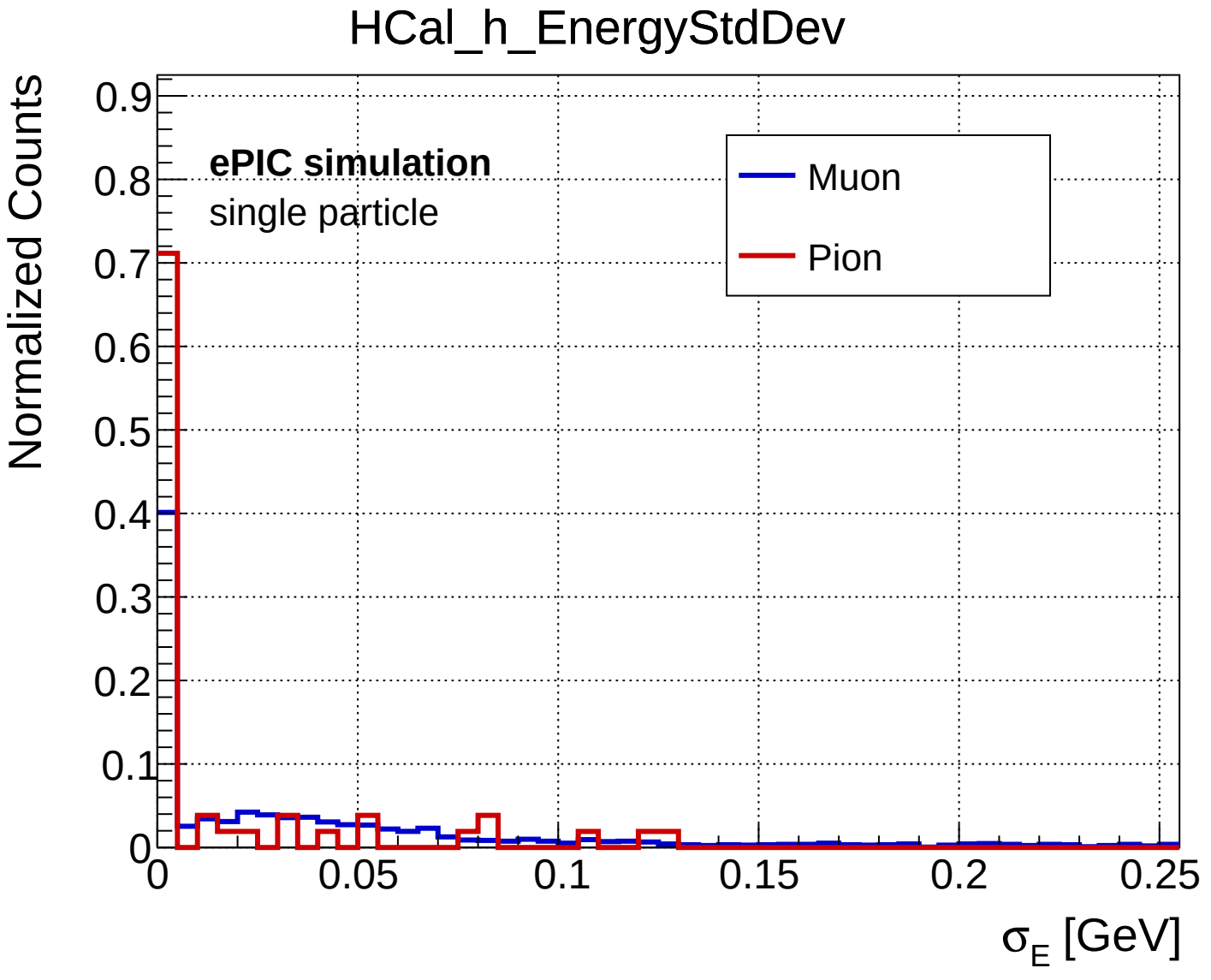

HCal_h_NHits
Normalized Counts
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
ePIC simulation
single particle
Muon
Pion
0
2
4
6
8
10
N hits

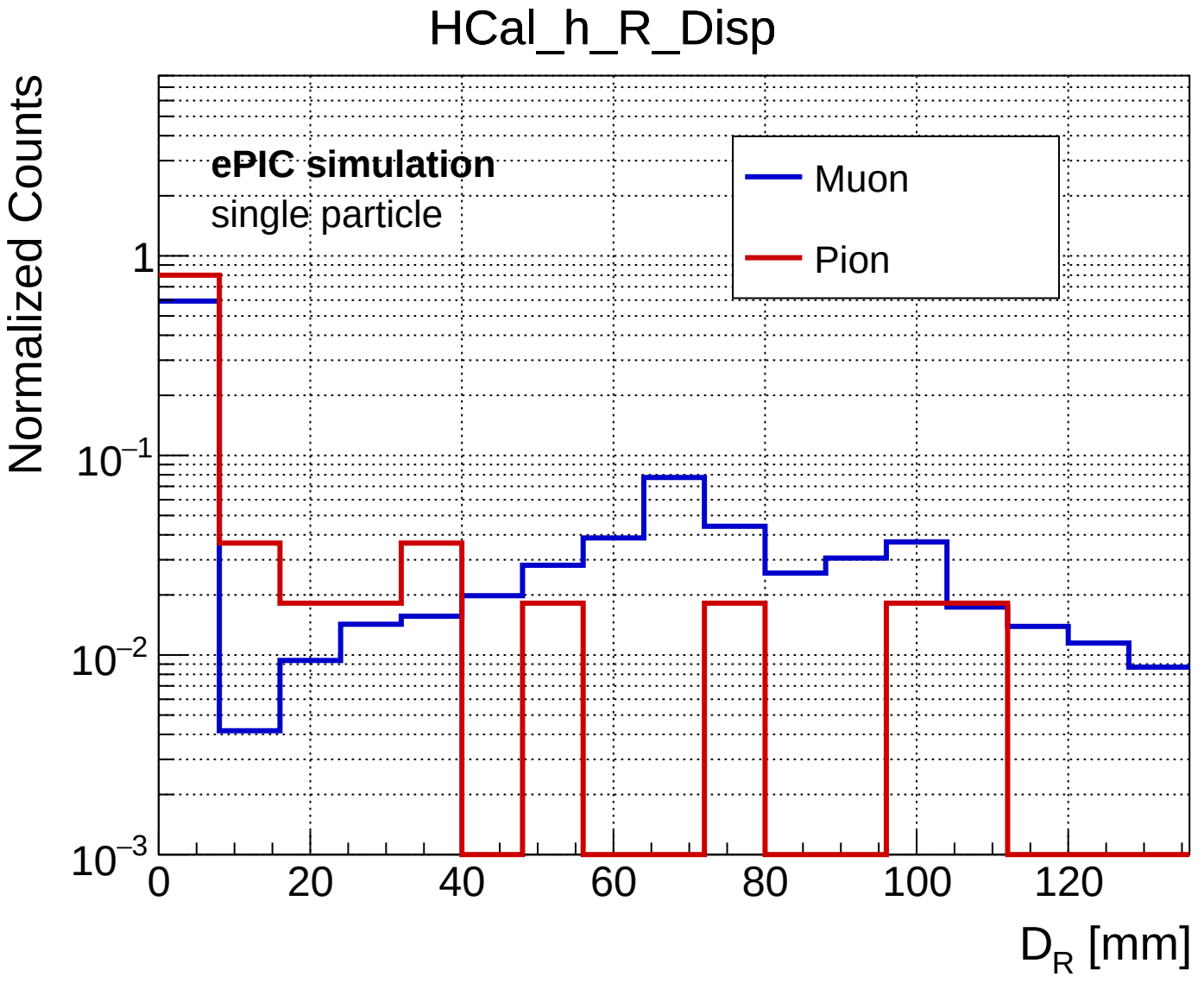

HCal_h_SpreadPhi
Normalized Counts
ePIC simulation
single particle
Muon
Pion
0
0.2
0.4
0.6
0.8
1
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
ϕ_{Spread} [rad]

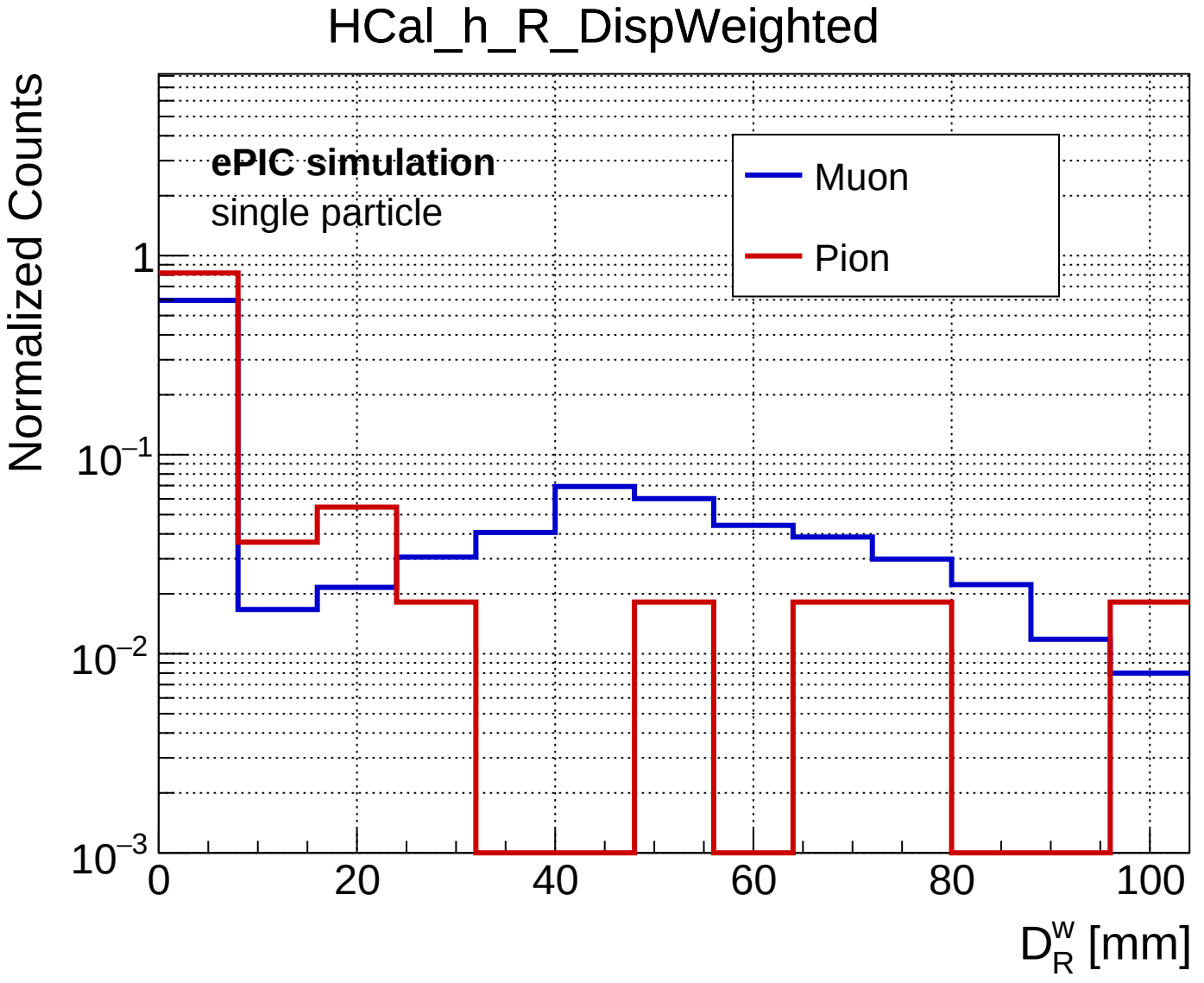

HCal_h_SpreadEta
Normalized Counts
ePIC simulation
single particle
Muon
Pion
0
0.2
0.4
0.6
0.8
1
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
0.22
η Spread

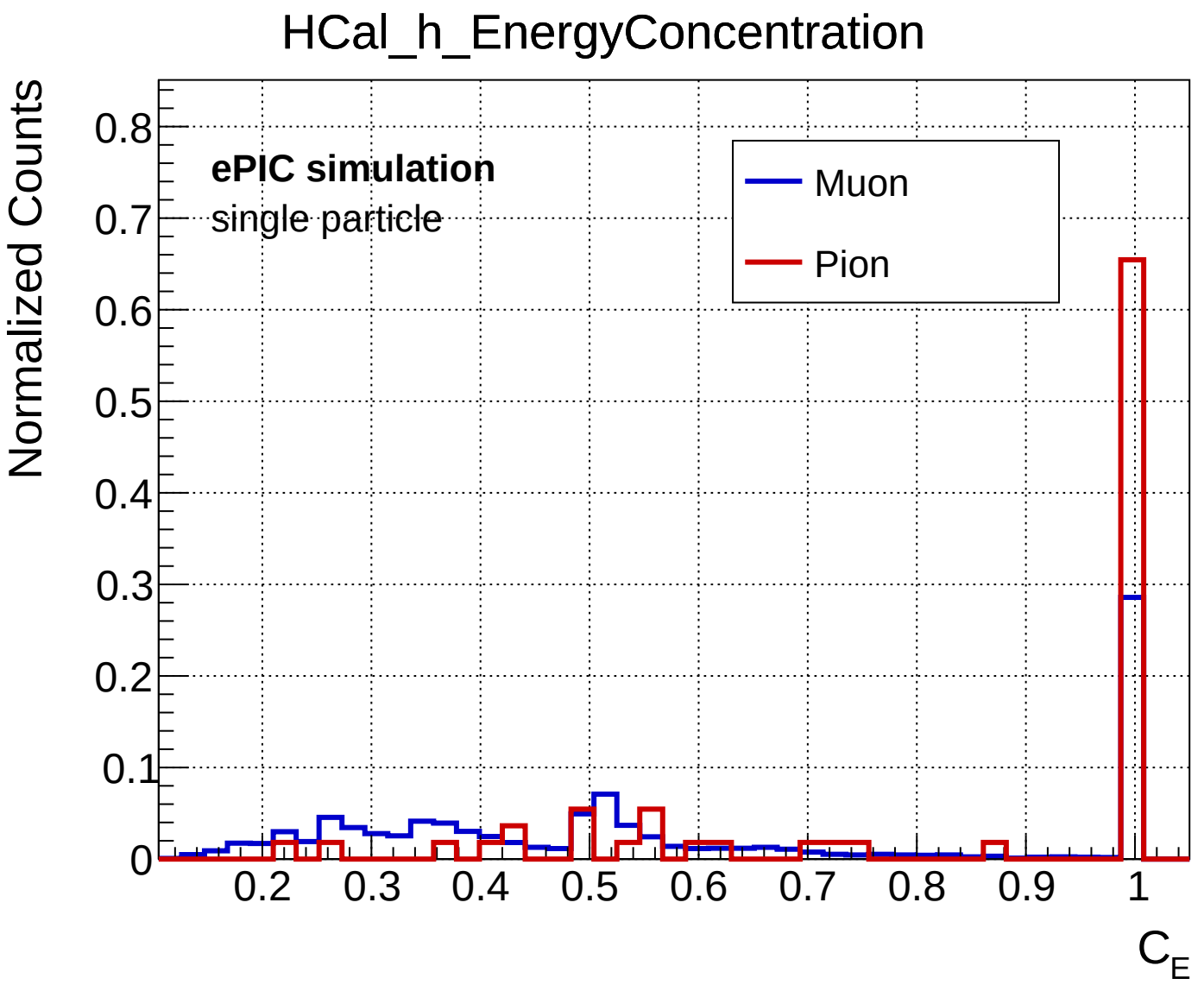

HCal_h_SpreadR
ePIC simulation
single particle
Muon
Pion
Normalized Counts
1
10^{-1}
10^{-2}
10^{-3}
0
50
100
150
200
250
300
350
R_{Spread} [mm]

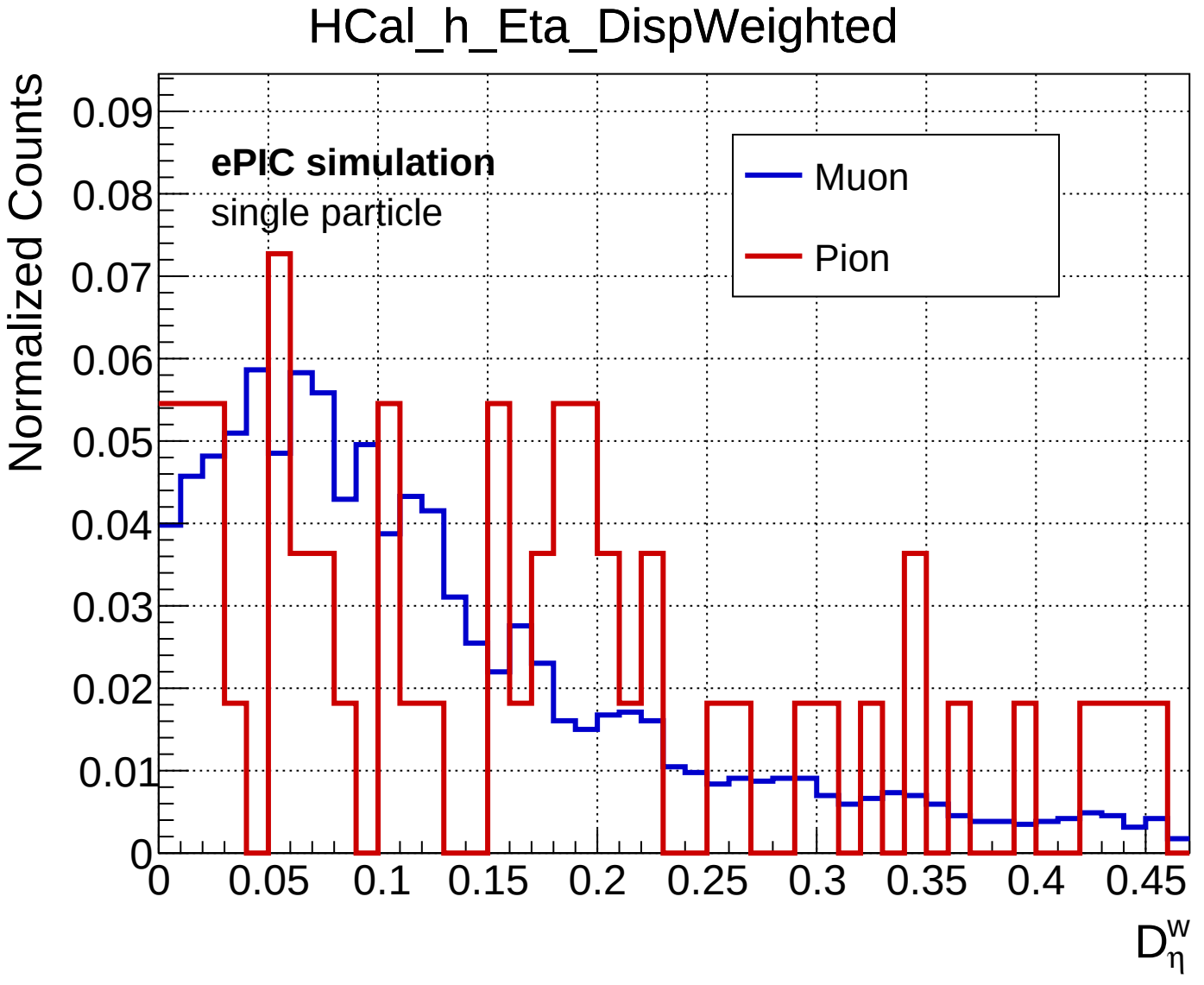

HCal_h_MaxHitFrac
Normalized Counts
ePIC simulation
single particle
Muon
Pion
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
f_max

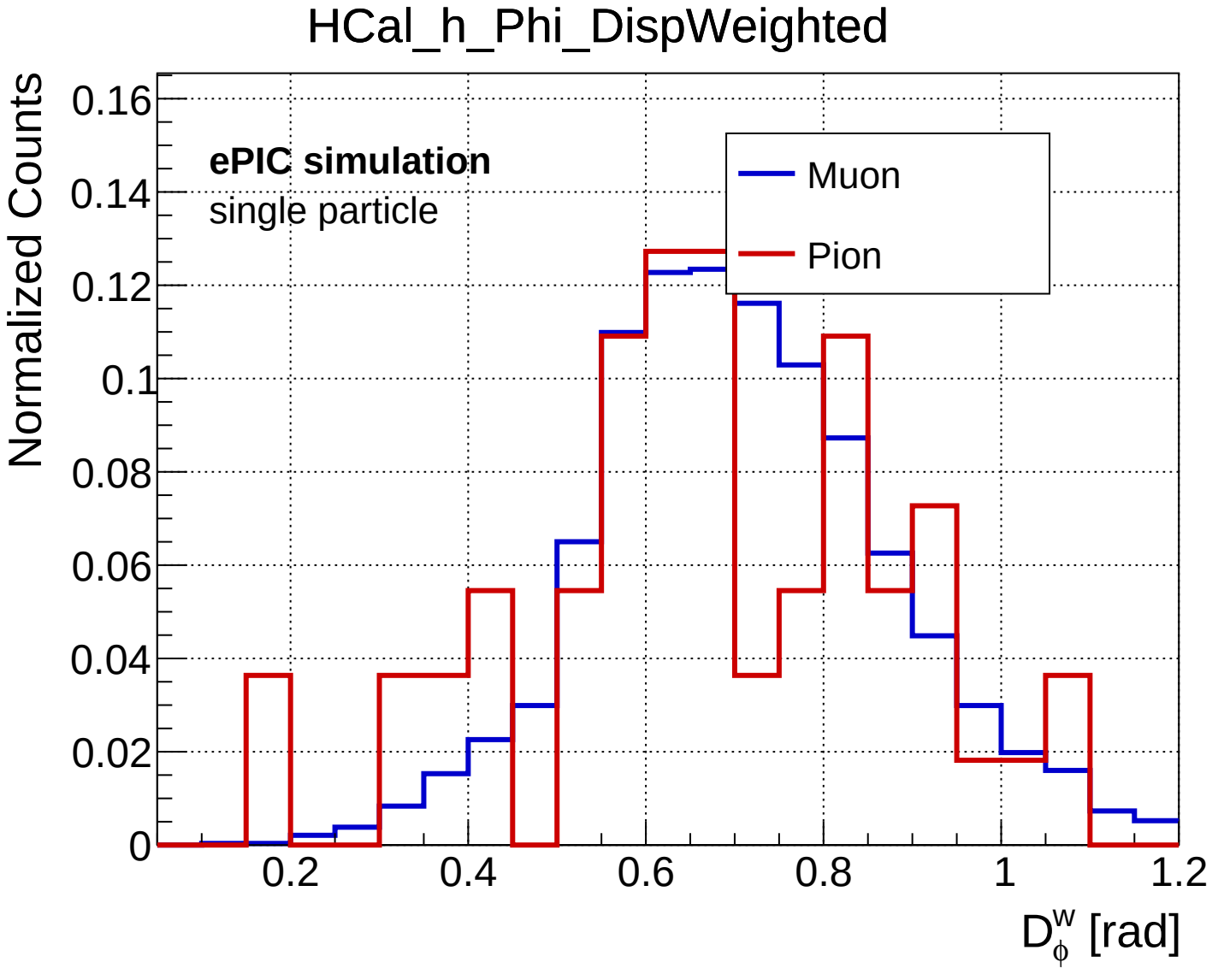

HCal_h_EnergyStdDev
Normalized Counts
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
ePIC simulation
single particle
Muon
Pion
0
0.05
0.1
0.15
0.2
0.25
σE [GeV]

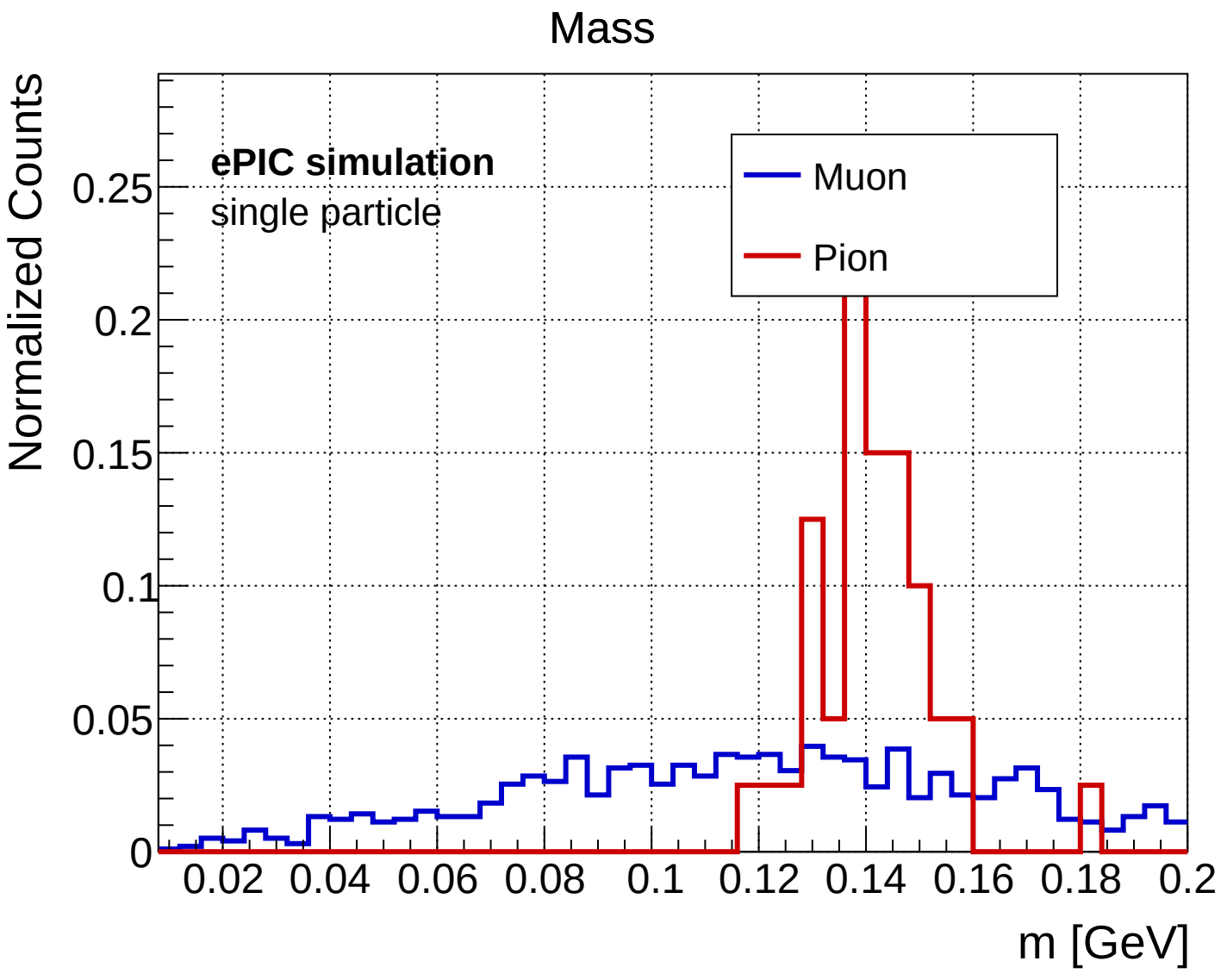

HCal_h_R_Disp
ePIC simulation
single particle
Muon
Pion
Normalized Counts
1
10^{-1}
10^{-2}
10^{-3}
0
20
40
60
80
100
120
D_R [mm]

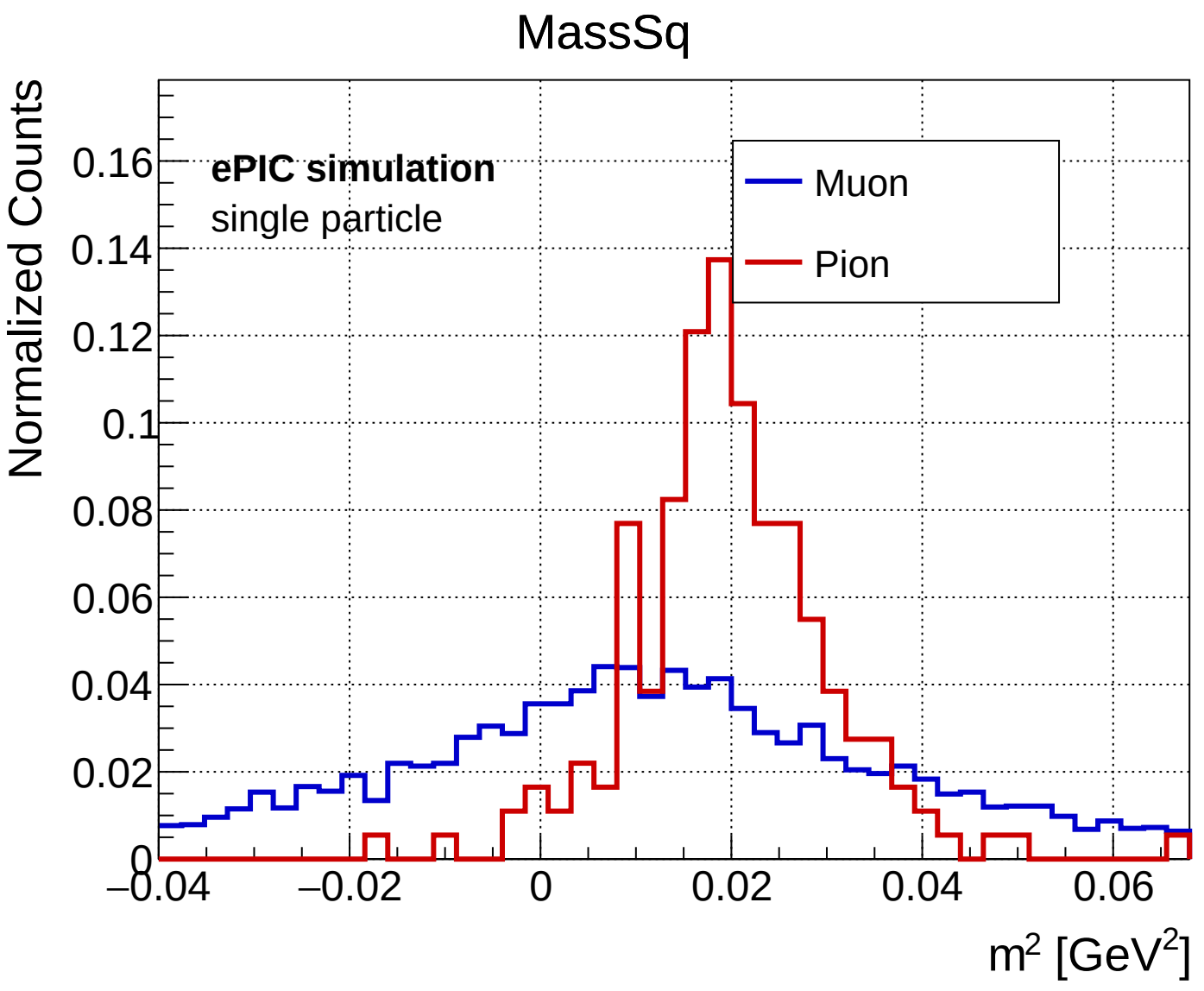

HCal_h_R_DispWeighted
ePIC simulation
single particle
Muon
Pion
Normalized Counts
1
10^{-1}
10^{-2}
10^{-3}
0
20
40
60
80
100
D_R^w [mm]

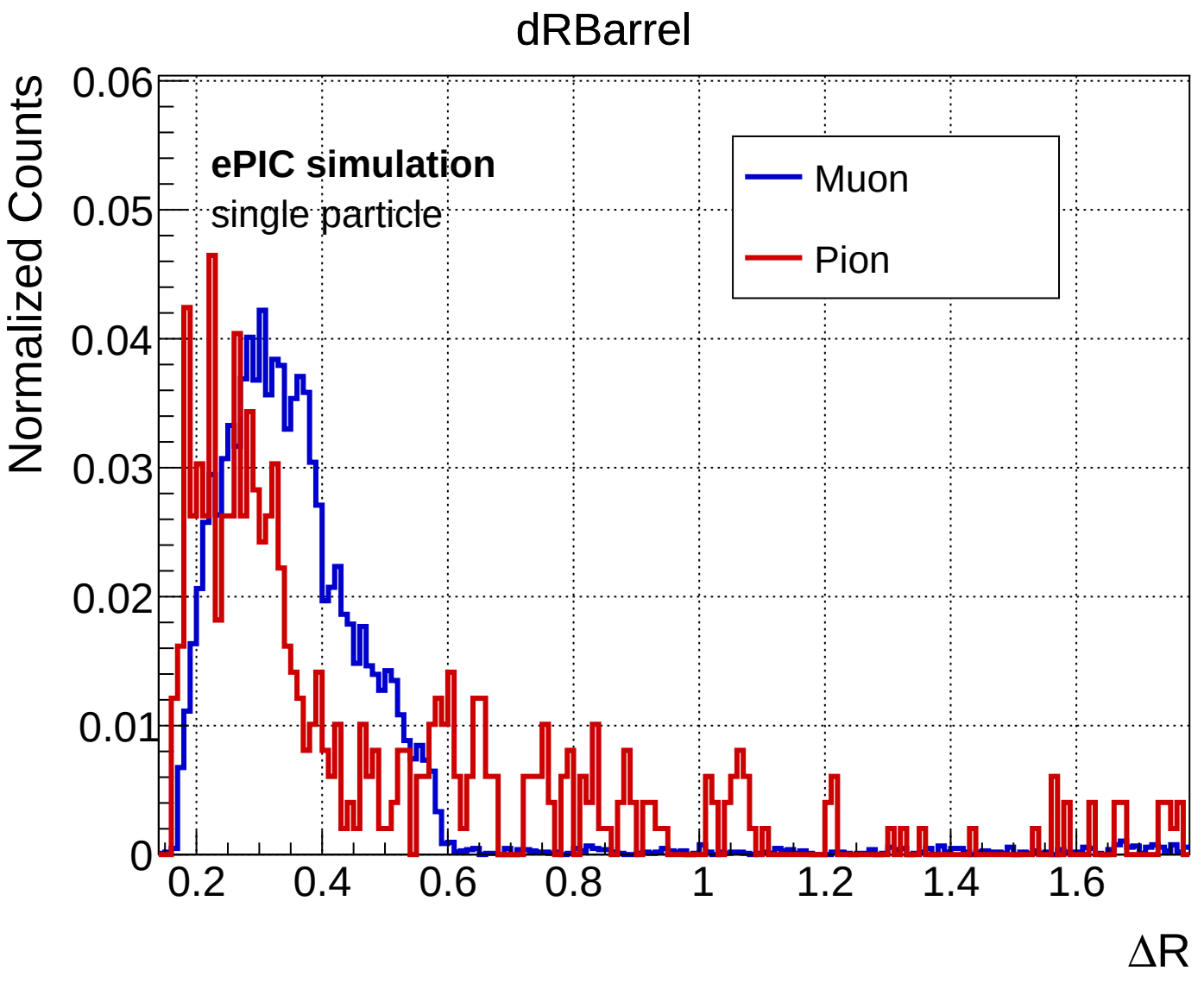

HCal_h_EnergyConcentration
Normalized Counts
ePIC simulation
single particle
Muon
Pion
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
C_E

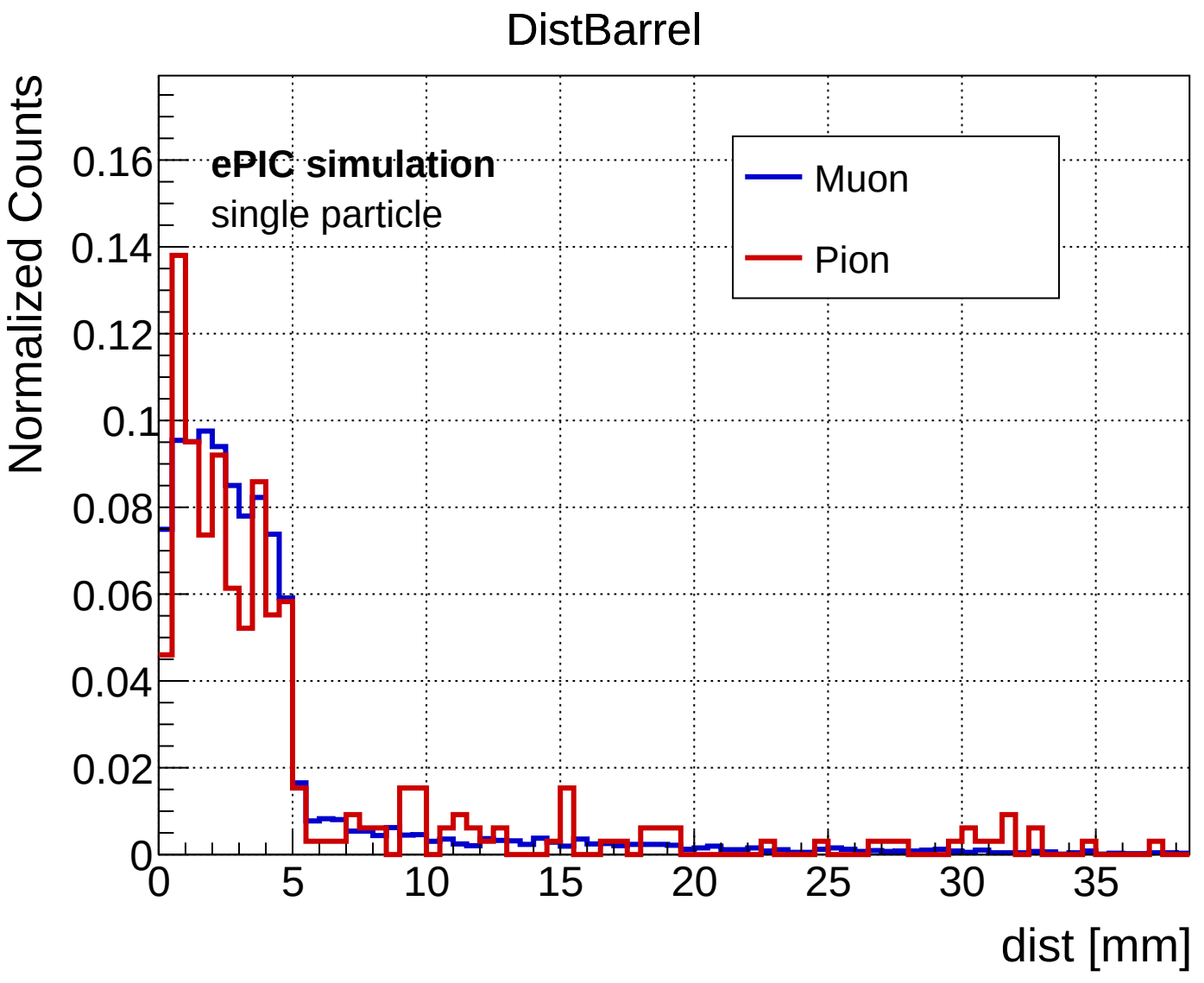

HCal_h_Eta_DispWeighted
Normalized Counts
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
ePIC simulation
single particle
Muon
Pion
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
D_η^w

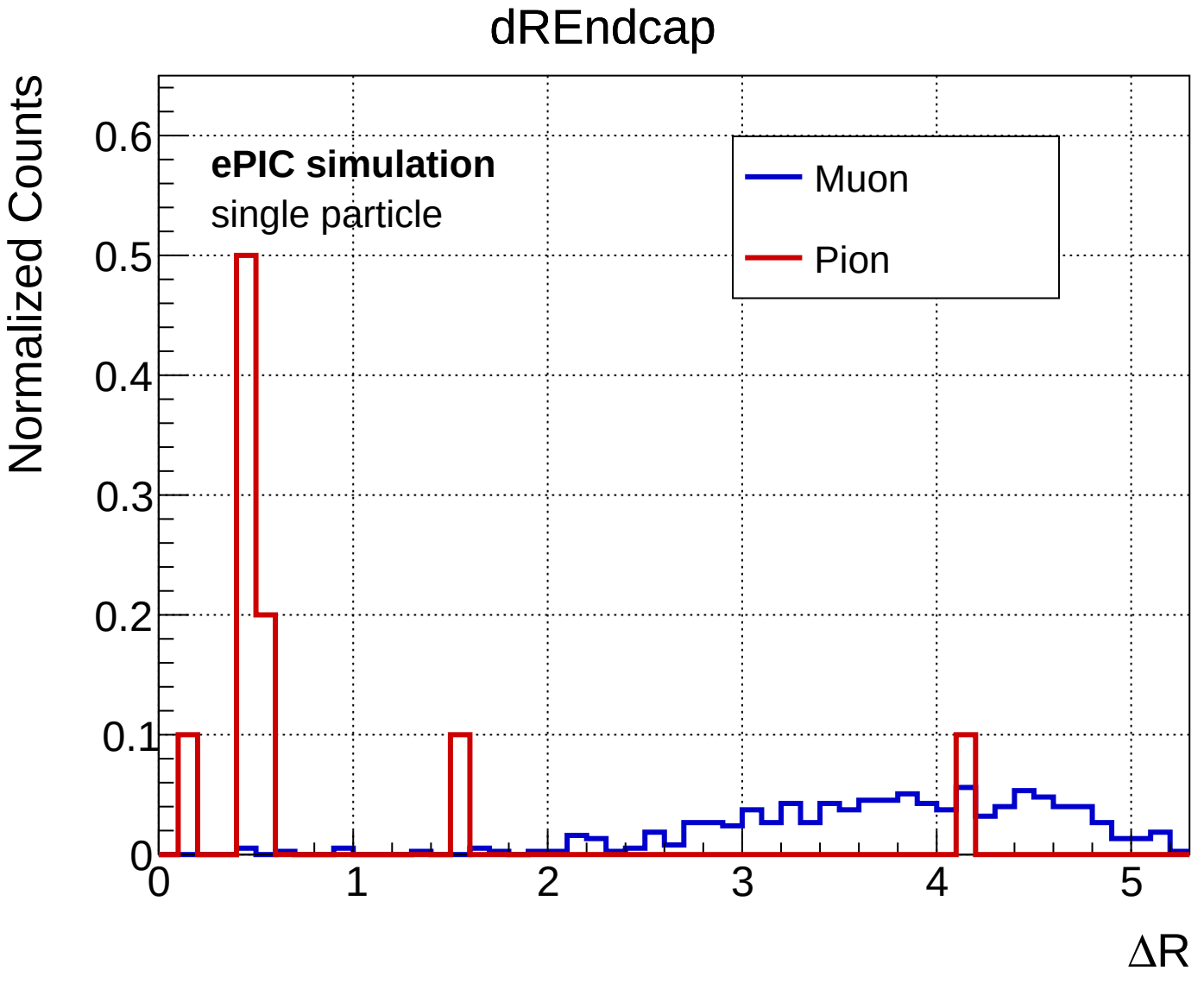

HCal_h_Phi_DispWeighted
Normalized Counts
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
ePIC simulation
single particle
Muon
Pion
0.2
0.4
0.6
0.8
1
1.2
D_{ϕ}^{w} [rad]

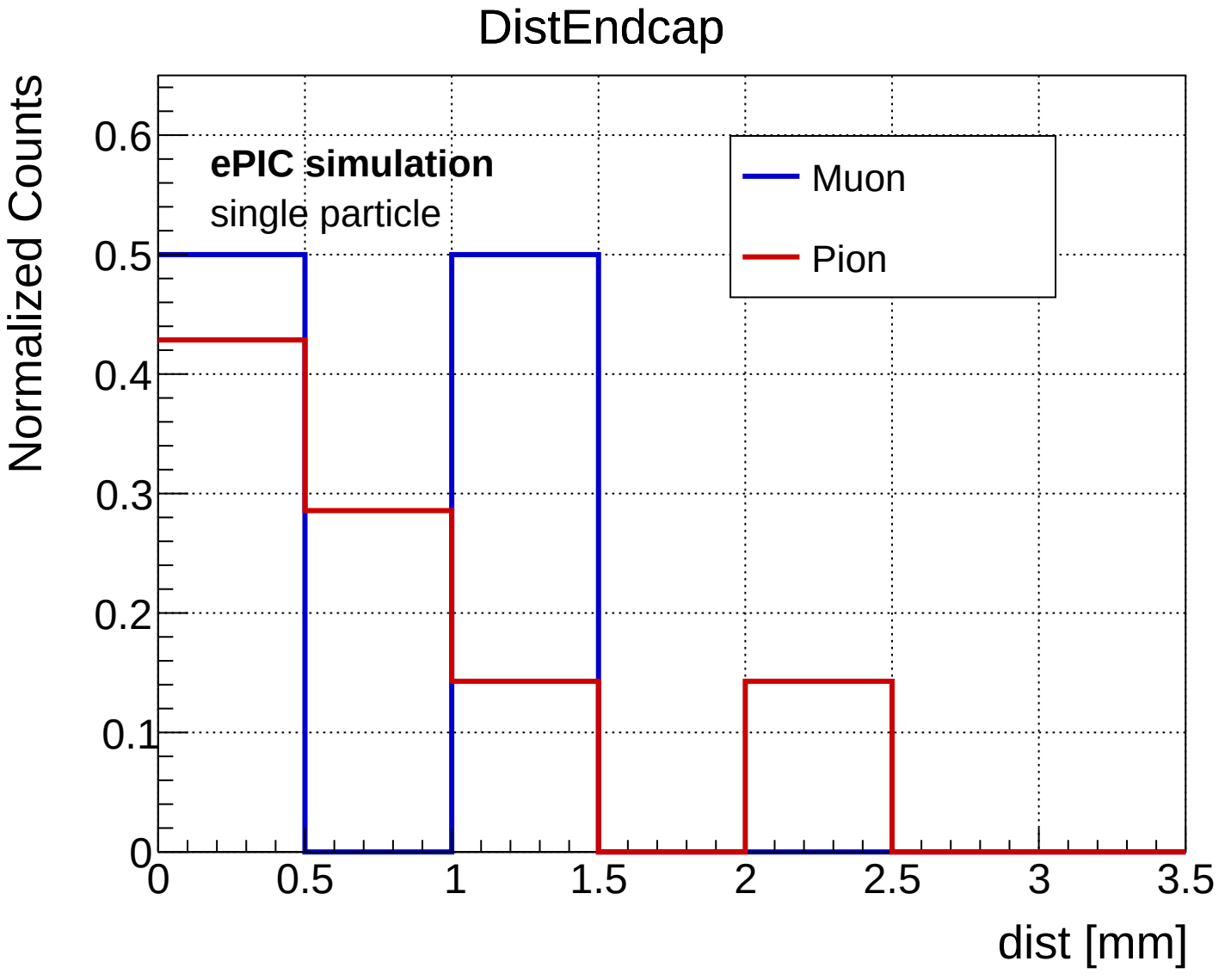

Mass
ePIC simulation
single particle
Muon
Pion
Normalized Counts
0
0.05
0.1
0.15
0.2
0.25
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
m [GeV]


MassSq
Normalized Counts
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
ePIC simulation
single particle
Muon
Pion
–0.04
–0.02
0
0.02
0.04
0.06
m² [GeV²]


dRBarrel
Normalized Counts
0.06
0.05
0.04
0.03
0.02
0.01
0
ePIC simulation
single particle
Muon
Pion
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
ΔR


DistBarrel
Normalized Counts
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
ePIC simulation
single particle
Muon
Pion
0
5
10
15
20
25
30
35
dist [mm]


dREndcap
Normalized Counts
0.6
0.5
0.4
0.3
0.2
0.1
0
0
1
2
3
4
5
ΔR
ePIC simulation
single particle
Muon
Pion


DistEndcap
Normalized Counts
0.6
0.5
0.4
0.3
0.2
0.1
0
ePIC simulation
single particle
Muon
Pion
0
0.5
1
1.5
2
2.5
3
3.5
dist [mm]